scientific reports

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Resilient federated intrusion detection with explainable AI: a robust CNN-LSTM architecture for extreme non-IID data distributions

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The rapid proliferation of Internet of Things (IoT) devices has led to a significant rise in security issues, as traditional centralized intrusion detection systems face difficulties in handling issues such as privacy concerns, communication bottlenecks, and heterogeneous data. Federated Learning (FL) is a new paradigm for collaborative learning that can be used to train intrusion detection systems without compromising user privacy. However, it is challenged by critical issues such as handling highly non-independent and identically distributed (non-IID) data, a common problem in heterogeneous IoT networks such as healthcare networks, financial networks, and industrial networks. In this paper, a novel CNN-LSTM architecture is proposed that is equipped with Explainable AI (XAI) to handle extreme cases of heterogeneous data in Federated Learning-based intrusion detection. Using the CIC-IDS2017 dataset and Dirichlet-based partitioning ($\alpha \in \{0.01, 0.1, 0.5, 5.0, 50.0\}$), our optimized CNN-LSTM model achieves a 97.36% centralized F1-score. We demonstrate that while architectural design provides highly stable robustness under moderate data heterogeneity ($\alpha = 0.5$), extreme non-IID conditions ($\alpha = 0.01$) trigger severe weight washing, dropping FedAvg performance to 71.53%. Through a comprehensive hyperparameter sweep, we prove that applying FedProx with a strong proximal penalty ($\mu = 0.1$) successfully mitigates this client drift, recovering the F1-score to 78.27%. Using SHAP, LIME, t-SNE, and PCA, we reveal that our model learns universal protocol-level features (Init_Win_bytes_forward, ACK Flag Count, and Fwd Packet Length Min) that remain invariant across heterogeneous networks. Finally, we demonstrate a highly stable detection is achieved for network-layer attacks, whereas application-layer intrusions (Web Attacks) suffer severe degradation, proving the fundamental limitations of flow-based features for payload-driven attacks.

Background and motivation

The proliferation of Internet of Things (IoT) devices, projected to surpass 31 billion active connections by 2030[2], has transformed the digital world in all sectors[1,2]. Smart cities now have millions of IoT devices that coordinate traffic lights, utilities, and emergency services. Autonomous vehicles combine camera feeds, LiDAR signals, and V2X communications to successfully traverse complex streets. Industrial control systems ensure that industrial processes like manufacturing and power production operate efficiently. Personal health monitoring devices continuously monitor vital signs and send the results to electronic health records in real-time. Such widespread connectivity has enabled many powerful features while also creating an enormous attack surface that is incredibly complex.

The security aspects of the connected Internet of Things are of considerable importance. The Mirai botnet attack in 2016 showed how insecure connected IoT devices could be hijacked to launch attacks on a global scale[3]. The 2015–2016 attack on the Ukrainian power grid using Supervisory Control and Data Acquisition system vulnerabilities resulted in power outages affecting hundreds of thousands of people[4]. These attacks show that the security of connected IoT devices goes beyond technology to include safety and security issues.

Traditionally, intrusion detection systems have been tested and have shown promise in traditional enterprise networks but have inherent issues in the context of IoT networks. The traditional centralized intrusion detection system follows a paradigm of 'collect and analyze,' where all data from connected devices is sent to a central server to analyze and detect possible attacks. The data is then fed into machine learning algorithms to detect possible attacks. The process seems straightforward but has inherent issues in the context of IoT networks. The

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first issue is the communication overhead. The smart city environment, where thousands of devices monitor different parameters, shows how sending data to a central point of analysis becomes a bottleneck.

Such a scenario becomes a major problem for a centralized IDS. The other major problem with a centralized IDS architecture is the occurrence of a single point of failure. The central server of a conventional IDS becomes a major threat to the overall security of the network. In the event of a malicious or non-malicious outage of the central server, the IDS becomes useless[5]. In critical infrastructure environments, this becomes a potentially life-threatening scenario. In environments where power plants, water treatment plants, and hospitals are present, the outage of a central server becomes a critical threat.

Another major problem with a conventional IDS architecture in an IoT environment is the threat of privacy and regulatory compliance. The HIPAA, GDPR, and PCI-DSS regulations place severe restrictions on data localization[6]. The recent €1.2 billion fine imposed by the GDPR on Meta Platforms Ireland for the unauthorized transfer of user data to the United States also highlights the severity of the threat of regulatory non-compliance. In a conventional IDS architecture, the threat of regulatory non-compliance and data privacy becomes a double-edged sword.

Machine learning for intrusion detection

Intrusion Detection Systems (IDS) based on Machine Learning have been proposed to address the fundamental limitations of Signature-based IDS, which cannot identify unknown or zero-day attacks without a pre-existing pattern database[7]. The advent of Deep Learning has revolutionized the IDS domain by allowing Behavioral IDS, rather than Signature-based IDS. Convolutional Neural Networks (CNNs) have been shown to effectively identify the spatial patterns in network packets, recognizing the divergence of attack traffic from legitimate traffic in terms of packet size, headers, and usage of different network protocols. Long Short-Term Memory (LSTM) networks have been used to identify the temporal patterns in the network traffic, allowing for the differentiation of slow port scans from legitimate traffic, thereby identifying the Advanced Persistent Threats (APTs) occurring over a period of time. A hybrid CNN-LSTM IDS has been shown to have achieved the state-of-the-art performance on standard datasets such as NSL-KDD, CICIDS2017, and CICIDS2018[8].

However, the performance of Deep Learning IDSs critically depends on the availability of high-quality labeled datasets. In the case of Distributed IoT networks, the data would remain scattered for several reasons, such as competitive, regulatory, or security concerns. For instance, a hospital would not share its patient network traffic, a financial organization would not share its transaction patterns, or a manufacturing organization would not share its production network traffic. However, the benefits of collective intelligence would accrue to all the organizations. For instance, the detection of a new variant of ransomware at a hospital would immediately provide security benefits to all the hospitals in the federation. This situation creates a fundamental tension between the benefits of localized data and the need for collaboration in the interest of security.

Federated learning: promise and challenges

Federated Learning (FL), which was formally introduced by McMahan et al.[9] at AISTATS 2017, is an approach to collaborative model development without sharing actual data by using an iterative process of local updates and global aggregation. FL has some significant benefits for security-focused applications: no actual data sharing is required, communication is limited to model updates instead of actual data sharing, computation is distributed across edge devices to relieve server burden, and regulatory requirements can be met by localizing the data[10]. FL has already been successfully implemented in industrial settings by Google (Gboard), Apple (Siri), and several healthcare organizations[11]. However, FL has an inherent disadvantage in that it can lead to non-IID conditions in the training set. IID is violated in nearly all FL settings. Feature Distribution Skew is present because of the inherent differences in the type of traffic that is present in IoT settings: financial networks typically have HTTPS traffic patterns, while industrial settings have SCADA traffic with specific timing patterns[12]. In the context of FL-based IDS systems, the problem of Label Distribution Skew is exacerbated because large organizations typically suffer from DDoS attacks, while small IoT networks suffer from bot scanning. Furthermore, some clients might never see specific types of attacks[13]. Quantity Skew is also an issue in FL because of the large range of data sizes that can affect the performance of FedAvg in concealing patterns that can be observed in small-scale settings[14]. Finally, there is the problem of Temporal Drift because of the dynamic nature of the attack surface. In fact, Zhao et al.[15] demonstrated that even with 10% non-IID conditions, FL can suffer from 30 to 50% reduction in accuracy, which can render an IDS system completely ineffective in practical settings.

The role of explainable AI in federated security

Apart from the performance aspects, the deployment of the intrusion detection system also requires security analysts to understand the decisions made by the model. A system that identifies potential anomalies but cannot provide the reasons for its decisions is not very helpful to the security analyst, who cannot effectively prioritize the response to the potential intrusions, adjust the detection threshold, or ensure that the system is detecting appropriate patterns for the right reasons. Explainable AI fills this critical need with techniques such as SHAP[24], LIME[25], t-SNE[26], PCA[27], which provide post-hoc interpretability to the decisions made by the deep learning models. With federated learning, the role of XAI is not limited to providing insights into the decisions made by the model but also to providing evidence of the mechanisms by which the model is able to provide non-IID robustness.

Contributions

This study makes the following four main contributions:

- Empirical demonstration of architectural robustness: In this paper, we report an extensive evaluation of the CNN-LSTM architecture across the entire range of non-IID data distributions ($\alpha \in \{0.01, 0.1, 0.5, 5.0, 50.0\}$). The CNN-LSTM achieves 96.94% centralized accuracy (mean, 3 seeds). Under extreme non-IID conditions ($\alpha = 0.01$), FedAvg accuracy is $80.30\% \pm 0.07\%$, representing a 16.64% degradation from centralized. Performance recovers substantially at moderate heterogeneity ($\alpha = 0.5$: 96.22%), demonstrating that the architecture remains viable when data distributions are not maximally skewed.
- Challenge to aggregation complexity orthodoxy: multi-seed, multi-μ evaluation of FedProx shows that aggregation choice is highly μ-sensitive: at $\alpha = 0.01$, $\mu = 0.001$ causes catastrophic failure (3.11% accuracy), $\mu = 0.01$ is competitive with FedAvg (80.18%), and $\mu = 0.1$ outperforms FedAvg (84.71% vs. 80.22%). This underscores that the earlier single-μ claim was confounded, and that proper hyperparameter tuning restores FedProx competitiveness.
- Mechanistic XAI analysis of non-IID robustness: In this paper, we report a comprehensive application of Explainable AI techniques in federated deep learning for IDS, using SHAP, LIME, t-SNE, and PCA in a synergistic manner. The analysis reveals the CNN-LSTM's ability to learn universal, protocol-level features (Init_Win_bytes_forward, ACK Flag Count, and Fwd Packet Length Min, TCP flags), which remain discriminative in the presence of heterogeneous data distributions.
- Challenge characterization in the attack classes: In this paper, we report the first analysis of the performance of the FL-IDS in the different classes of attacks, showing exceptionally high performance in the case of network-layer attacks ($F1 > 0.99$).

The rest of this paper is organized as follows: In Section "Related work", a thorough overview of the theoretical underpinnings of Federated Learning (FL), the challenges associated with the issue of non-IID data, and the recent advancements in deep learning and explainability in the context of IDS will be provided. In Section "Methodology", the proposed methodology will be described in detail, including the five-stage data preprocessing pipeline, class balancing using the SMOTE technique, feature selection via the Chimp Optimization Algorithm (ChOA), and the explicit design of the hybrid CNN-LSTM model. In Section "Results and analysis", the experimental results will be provided, including the performance of the centralized model and the thorough evaluation of the federated model under various Dirichlet-distributed non-IID severities. Additionally, this section will include an evaluation of the performance of the two FL protocols, i.e., FedAvg and FedProx, to ensure the robustness of the architectural design. In Section "Discussion", the multilayered explainability analysis will be provided, including the use of SHAP, LIME, t-SNE, and PCA to shed light on the mechanistic underpinnings of the robust performance in the presence of non-IID data. In Section "Conclusion", the various factors affecting the robust performance in the presence of non-IID data will be provided, along with the challenges faced in the presence of application-layer attacks and the implications for practitioners and potential research directions. In the final section, the paper will be concluded by briefly describing the findings and contributions.

Related work

Federated learning foundations

The FedAvg algorithm was introduced by McMahan et al.[9], thereby laying the foundation for federated learning by iteratively performing local updates and averaging global model parameters. The convergence supports for IID conditions were provided by Li et al.[28] thereby proving that Federated Learning could be as effective as centralized learning by performing a large number of communication rounds and by tuning the learning rate. Yang et al.[10] provided a comprehensive framework for Federated Machine Learning by clearly describing Horizontal Federated Learning, Vertical Federated Learning, and Federated Transfer Learning.

Non-IID challenges in federated learning

Zhao et al.[15] gave a systematic quantification of non-IID degradation and found that accuracy is reduced by 30–50%. This work sparked a decade-long line of work in algorithmic mitigation techniques. Li et al.'s FedProx[16] proposed the use of proximal regularization to bound the variation in local models, SCAFFOLD[17] maintained client-side control variates to correct for client drift, increasing communication overhead by twofold, and FedNova[18] normalizes the weights to capture the effect of different local training frequencies and tackle the objective inconsistency problem in heterogeneous federated optimization, and Karimireddy et al.[29], which identified client drift as the main contributor to non-IID degradation.

Systematic works by Zhu et al.[30] and Hsieh et al.[31] proposed taxonomies of non-IID data, such as label skew, feature skew, quantity imbalance, and temporal drift, which show that non-IID data is a multi-faceted problem. More importantly, Moussaoui et al.[32] presented a systematic review of hybrid AI models that combine machine learning and FL and concluded that architectural solutions are under-explored in comparison to algorithmic solutions, which is what our work focuses on.

Deep learning for intrusion detection systems

Deep learning technologies have revolutionized the IDS domain, transforming it from signature-based to behavioral IDS. Javaid et al.[33] proved the superiority of deep sparse autoencoders over traditional machine learning techniques on the NSL-KDD dataset. Yin et al.[34] proposed the RNN-IDS, which achieved 93% accuracy on the KDD99 benchmark using LSTMs for sequence classification, thereby proving the existence of discriminative information in temporal dependencies within network flow data. Li et al.[35] also demonstrated the supremacy of 1-D CNNs over other machine learning techniques on the same benchmark, thereby proving the existence of discriminative information in raw packet byte data. Nguyen et al.[36] proposed the use of CNNs for traffic classification, achieving 98% accuracy on the CICIDS2018 benchmark, thereby proving the existence of discriminative information in raw packet byte data. Vinayakumar et al.[8] proposed the CNN-LSTM IDS, which

surpassed the accuracy of other machine learning techniques by 3–5% on the CICIDS benchmark, thereby validating the proposed IDS architectural principles.

The contemporary advancements represent the cutting-edge technology. Zhang et al.[37] proposed the GNN-IDS, which attained state-of-the-art results on multiple benchmarks by exploiting the discriminative information present in the topological relationship between network flow data. The network traffic is modeled as a graph, where each node represents the network flow, and the edges represent the characteristics shared by the network flow. Wang et al.[38] proposed the hierarchical attention network, which allows the network to concentrate on the relevant time segments by utilizing the attention mechanism. Lin et al.[39] proposed the ET-BERT, which attained 97.8% accuracy on the network traffic classification task by utilizing the attention mechanism, as proposed by Bahdanau et al.[55]. Despite the significant achievements, the federated deployability of these state-of-the-art architectures, especially in the extreme non-IID case, is largely unknown, thereby exposing the critical gap in the existing state-of-the-art. Furthermore, recent methodologies have successfully demonstrated the efficacy of feature-enhanced deep belief networks[57] and hybrid swarm-intelligent neural networks[58] for dynamic threat prediction, underscoring the growing importance of mathematically optimized feature spaces in cybersecurity.

Federated intrusion detection systems

Federated learning for IDS has evolved from the demonstration of feasibility to the development of specialized federated IDS. Nguyen et al.[40] presented the DIoT framework, which utilizes federated support vector machines for anomaly detection in IoT environments. Although the study demonstrated the feasibility of federated learning for IDS, the ability of support vector machines to handle high-dimensional nonlinear relationships, which are the norm in today's complex network traffic, is limited. Zhao et al.[41] presented the FedCNN-IDS framework, which achieved high accuracy of 91% on the NSL-KDD dataset. However, the study only tested the framework under mild non-identically distributed conditions ($\alpha > 1.0$). The study did not address the issue of extreme heterogeneity, which is a major challenge in federated IDS. Liu et al.[42] utilized the FedProx algorithm for federated IDS. The study achieved high accuracy of 89% under moderate non-identically distributed conditions ($\alpha = 0.01$). However, the study assumed the need for the complexity of the FedProx algorithm without testing the efficacy of a less complex aggregation algorithm when a robustly designed IDS architecture was utilized. This requirement for effective intrusion detection within complex CPS and IoT systems has created a need for architectures that are able to process highly dimensional data while maintaining computational efficiency. Recent literature has highlighted how the integration of meta-heuristic feature optimization with hybrid deep learning or ensemble-based models has a significant impact on the model's ability to identify minority class-based attack signatures within the network[20–22]. Drawing directly from existing optimization methodologies, the approach integrates the Chimp Optimization Algorithm (ChOA) as an offline pre-deployment approach to ensure that the CNN-LSTM model operates within a mathematically optimized feature space prior to being exposed to extreme non-IID data distributions.

Federated extensions for the purpose of ensuring the privacy of users have gained a lot of scholarly attention. Preuveneers et al.[43] presented the application of differential privacy for federated anomaly detection. The study achieved high levels of privacy guarantee of ($\varepsilon = 3.0$, $\delta = 10^{-5}$) at a cost of 7% in terms of accuracy. Ahmed et al.[44] presented a federated learning framework integrated with explainable AI for intrusion detection. While the study achieved high detection rates, it primarily focused on standard feature spaces without optimizing for extreme Dirichlet-based non-IID data distributions. Similarly, recent studies have explored hybrid bagging and boosting ensembles[45] to improve the detection of complex intrusions. However, deploying these complex ensembles in resource-constrained federated environments remains computationally challenging, highlighting the need for lightweight, inherently robust architectures. More recently, Wang and Liu[50] demonstrated federated learning-based intrusion detection for industrial IoT networks, achieving competitive detection rates while preserving data locality across heterogeneous industrial clients.

Explainable AI for network security

XAI in security applications has started to gain traction with the migration of deep learning-based intrusion detection systems (IDS) into production environments that require operator oversight. SHAP[24] provides theoretically sound explanations based on cooperative game theory that satisfy consistency, missingness, and efficiency axioms. LIME[25] provides explanations for individual predictions through local linear model approximations that provide intuitive explanations for non-technical security analysts. Dimensionality reduction techniques like t-SNE[26] and PCA[27] provide visualization of the learned feature space geometry that allows for validation that learned features are universal rather than client-specific. XAI for federated learning is still in its infancy; however, in this work, we provide a comprehensive multi-method XAI analysis for federated deep learning-based IDS.

Benchmark datasets for IDS research

The quality of IDS research is heavily dependent on the realism of the benchmark data sets. While the KDD Cup '99 and NSL-KDD data sets[46,47] represent the attack patterns of the 1990s, along with the unrealistic traffic models, the CICIDS2017[59] dataset, developed by the Canadian Institute of Cybersecurity, features real background traffic and 14 contemporary attack classes, making it the gold standard today, with 2,827,876 labeled data flows, 78 features extracted by CICFlowMeter, five days, and simulation of the network environment. Extensions like CICIDS2018 and CIC-DDoS2019 extend the scope of the attack classes. CICIDS2017 is the most popular data set for federated learning research, given its comprehensiveness, publicly available nature, and the presence of heterogeneous traffic patterns over multiple days, which is required for federated learning to offer the required attack class diversity.

Summary of related works and research gap

In addition, a comprehensive quantitative and qualitative comparison of our proposed XAIFIDS framework against these state-of-the-art methods is presented in Sect. "Comparative discussion" (Table 5), comparing it to the state-of-the-art in federated IDS systems. Indeed, some of the early accomplishments in the field, such as the ones proposed in Zhao et al.[41], Liu et al.[42], demonstrated the viability of temporal architectures, such as CNN-LSTM, although these early approaches faced issues related to mild to moderate data heterogeneity and lack interpretability layers. On the other hand, some of the very recent state-of-the-art approaches from 2025 and 2026, such as Fatema et al.[51], Alhayan et al.[53], incorporated XAI techniques such as SHAP and LLM explanations, although these approaches still assume IID scenarios or utilize simplistic partitioning schemes. Our proposed XAIFIDS framework is timely and resolves a major gap in the state-of-the-art in the following ways: Our proposed XAIFIDS framework is a comprehensive application to concurrently address all three aspects, which are normally treated individually in the state-of-the-art, namely architectural robustness under extreme Dirichlet-based non-IID scenarios ($\alpha \leq 0.01$), the multi-algorithm federated framework, and the synergistic XAI suite.

Methodology

Figure 1 illustrates the comprehensive architecture of the proposed resilient federated intrusion detection framework. At the edge, individual clients preprocess local, non-IID traffic data using SMOTE balancing to mitigate class imbalance prior to training. The foundation of the learning process is a hybrid CNN-LSTM architecture. This model leverages CNN modules to isolate hierarchical spatial features, while subsequent LSTM layers capture sequential temporal dependencies. Crucially, by focusing on universal representations at the protocol level, the architecture is specifically engineered to maintain structural robustness and high performance even under extreme non-IID data conditions. Following local training, model updates are transmitted to a central Federated Learning Server and aggregated using the FedAvg algorithm. Finally, to ensure transparency for security analysts, an integrated Explainable AI (XAI) module employs SHAP and LIME to provide both global feature importance and local instance-level explanations.

Dataset selection and preprocessing

For performing the experimentally-driven evaluation, the CICIDS2017 dataset is used. Despite the recent emergence of even newer datasets in literature, CICIDS2017 has become a de facto benchmark for evaluating intrusion detection systems built based on Federated Learning approaches. The use of CICIDS2017 is well justified by its great variety, which involves 2,827,876 samples and 14 network attack types, created during 5 days of background activity. This level of diversity becomes essential for evaluating the effectiveness of our proposed approach in the case of extremely challenging Dirichlet-based partitioning with non-IID distribution ($\alpha \leq 0.01$), since it allows us to generate a sufficient number of minority class samples, required to simulate highly non-IID IoT devices' behavior. The dataset has 78 features created by the CICFlowMeter tool, including flow-based, packet-based, flag-based, and time-based features. In order to assure the methodology of our approach, and to avoid any form of data leakage between federated clients, we define the five mathematically-defined stages of the preprocessing procedure.

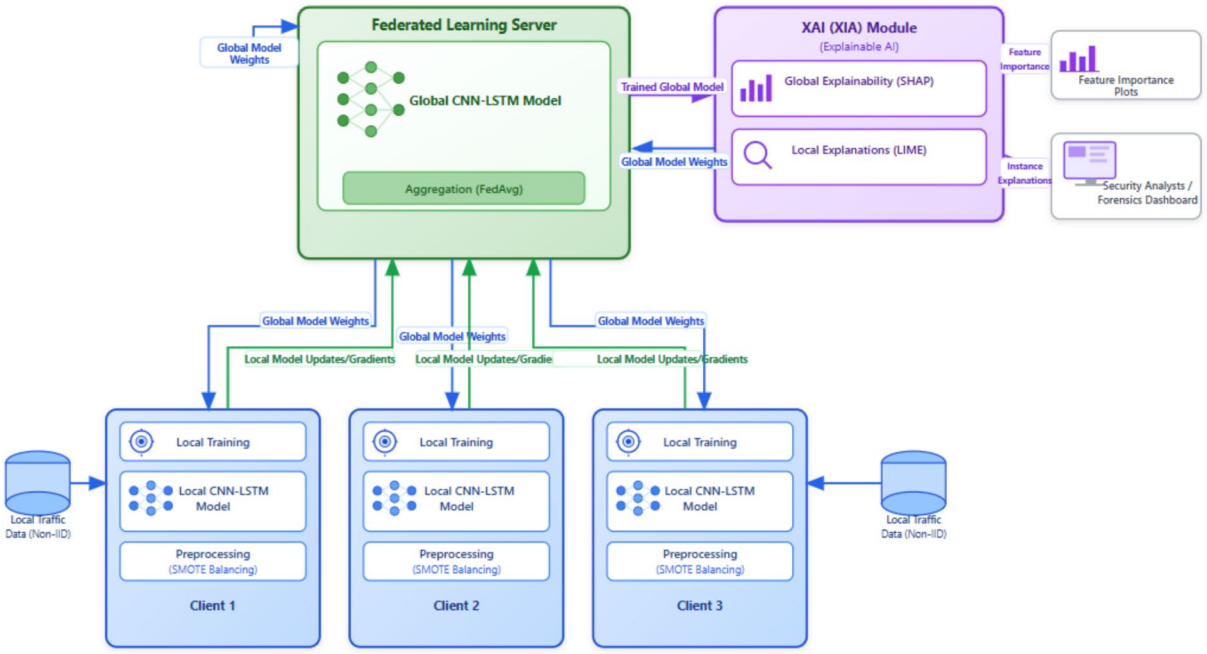


Fig. 1. XAIFIDS model architecture.

Step 1: Stratified Sampling. To perform the experiments within the computational bounds while maintaining statistical representativeness, stratified sampling is used at a ratio of $p = 0.05$:

$$\mathcal{D}_{sample} = StratifiedSample\left(\mathcal{D}_{orig}, p = 0.05\right) \tag{1}$$

The stratified procedure ensures class proportion preservation: for each class $c \in C$,

$$\frac{|\{(x_i, y_i) \in \mathcal{D}_{sample} : y_i = c\}|}{|\mathcal{D}_{sample}|} \approx \frac{|\{(x_i, y_i) \in \mathcal{D}_{orig} : y_i = c\}|}{|\mathcal{D}_{orig}|} \tag{2}$$

These yields $|D_{sample}| = 141{,}394$ samples preserving the original distribution across all 14 classes. To preserve the temporal integrity of network attacks, this sampling is executed strictly at the flow-session level, ensuring that multi-step attack sequences remain chronologically coherent for realistic sequential evaluation.

Step 2: Class Filtering. For each class c, define class support $n_c = |(xi, yi) \in D_{sample} : yi = c|$. We remove classes with insufficient support below minimum threshold $\tau = 6$:

$$\mathcal{C}_{filtered} = \{c \in \mathcal{C} : n_c \geq \tau = 6\} \tag{3}$$

Classes removed: Infiltration *(n = 3)*, Web Attack SQL Injection *(n = 4)*, Heartbleed *(n = 0)*. This yields 13 classes (12 attack categories + BENIGN).

Step 3: Handling Infinite and NaN Values. The CICFlowMeter feature set contains infinite values, which result from division by zero in the flow rate calculation. These values are systematically handled:

$$x_{ij} \leftarrow \max_k\left(x_{kj}\right) if x_{ij} = +\infty, \qquad x_{ij} \leftarrow 0 if x_{ij} = NaN \tag{4}$$

This ensures numerical stability without discarding affected samples, which would introduce selection bias.

Step 4: Min–Max Feature Normalization. For each feature $j \in 1,2,\ldots,78$, Min–Max normalization maps all features to [0,1], ensuring uniform contribution to gradient computation:

$$x_{ij}^{norm} = \frac{x_{ij} - \min_k\left(x_{kj}\right)}{\max_k\left(x_{kj}\right) - \min_k\left(x_{kj}\right)} \in [0,1] \quad \forall i, j \tag{5}$$

Critical: Normalization parameters (min_j, max_j) are computed exclusively on the training partition and applied identically to validation and test sets, preventing data leakage that would inflate reported test performance.

Step 5: Label Encoding. String class labels are mapped to integer indices through a consistent, reproducible mapping:

$$y_i^{enc} = LabelEncoder\left(y_i\right), \quad y_i^{enc} \in \{0,1,\ldots,12\} \tag{6}$$

followed by one-hot encoding for multi-class categorical cross-entropy training:

$$y_i^{ohe} = OneHot\left(y_i^{enc}\right) \in \{0,1\} \tag{7}$$

To simulate extreme data heterogeneity across the federated edge network, the training data is partitioned among *K* clients using a Dirichlet distribution. For a dataset with *C* classes, we sample the class prior distribution p_k for each client *k* as follows:

$$p_k \sim Dir\left(\alpha q\right) \tag{8}$$

where *q* is the global prior class distribution, and $\alpha > 0$ is the concentration parameter controlling the degree of non-IID skew. A smaller α (e.g., $\alpha = 0.01$) results in extreme label imbalance, forcing individual clients to hold data from only one or two classes, accurately reflecting the localized, siloed nature of IoT sensor deployments.

SMOTE-based class balancing

The data set shows a clear class imbalance: BENIGN traffic constitutes about 83% of the data set, whereas other classes of attacks occur less than 100 times in the corpus. The problem with training data with this clear imbalance is that the classifier will be skewed towards the majority classes. In the context of intrusion detection, where the recall of minority classes (detection rate) is of critical importance, this is undesirable.

To overcome this problem, we use the SMOTE (Synthetic Minority Over-sampling Technique) algorithm[19]. Crucially, to prevent data leakage and ensure practical deployability, SMOTE is strictly executed locally and independently on each client's isolated training partition (D_{train}^k) only after the Dirichlet distribution has assigned the data, where synthetic data points are created for minority classes. The algorithm for the SMOTE technique, for a minority class c where $n_c < n$ target, is as follows:

Input: X_c (minority class samples); k = 5 (nearest neighbours); N_{target} (desired count)

Output: $X_c^{balanced}$ — augmented dataset where $|X_c^{balanced}| = N_{target}$

Init: $X_c^{balanced} \leftarrow X_c$

while $|X_c^{balanced}| < N_{target}$ do

  Select random sample x_i from X_c

  Neighbor search: compute $N_k(x_i)$ via $d(x_i, x_j) = |x_i - x_j|_2$

  Neighbor selection: draw $x_r \sim Uniform(N_k(x_i))$

  Interpolation: $x_{syn} = x_i + \lambda(x_r - x_j),\ \lambda \sim Uniform(0, 1)$

  Label: $y_{syn} \leftarrow c$

  Update: $X_c^{balanced} \leftarrow X_c^{balanced} \cup \{(x_{syn}, y_{syn})\}$

end while

Constraint: applied only to $\mathcal{D}_{train}$ — prevents leakage into $\mathcal{D}_{test}$

return $X_c^{balanced}$

Algorithm 1. SMOTE-based class balancing.

Result: Dataset expands from 141,394 to 1,360,608 samples (9.6 × expansion), with all 13 classes balanced to approximately 104,662 samples each.

To ensure strict evaluation integrity, SMOTE is applied exclusively after the global train/test split, ensuring the hold-out test set contains absolutely zero synthetic data. Furthermore, for the federated training pipeline, SMOTE is applied to the global training pool prior to Dirichlet partitioning. This is a necessary simulation design choice; under extreme non-IID skew (α = 0.01), individual clients often receive zero instances of minority attack classes, which would render local post-split SMOTE mathematically impossible. Pre-partition balancing ensures these critical minority signatures survive the distribution process to properly test the robustness of the federated aggregation algorithms.

Feature selection via chimp optimization algorithm

The original 78 features include many redundant or irrelevant dimensions. We apply ChOA[49] to find the optimal feature subset $F* \subseteq 1, \ldots, 78$ maximizing the objective:

$$F^* = \arg \max_{F \subseteq \{1,\ldots,78\}} \left[Accuracy\left(M\left(X_F, y\right)\right) - \alpha \cdot \frac{|F|}{d} \right] \tag{9}$$

where α = 0.01 penalizes subset size. ChOA maintains $N_{pop} = 30$ binary candidate vectors over $T_{max} = 50$ iterations, using four leader chimps (Attacker, Barrier, Chaser, Driver) to guide population updates. Binary conversion via sigmoid thresholding determines feature inclusion.

17 Features Selected: Init_Win_bytes_forward, ACK Flag Count, and Fwd Packet Length Min, Init_Win_bytes_forward, PSH Flag Count, Destination Port, Down/Up Ratio, ACK Flag Count, Bwd Packet Length Max, Max Packet Length, Packet Length Mean, Fwd IAT Std, Fwd Packet Length Mean, URG Flag Count, Fwd IAT Max, Bwd IAT Total, Bwd Packet Length Mean, Average Packet Size—a 78% dimensionality reduction.

By identifying these 17 features a priori, we establish a lightweight, optimized input vector for the CNN-LSTM architecture, ensuring that no private local client data is accessed or leaked to the server during the feature selection phase.

Note that this ChOA feature profiling is conducted strictly as an offline, pre-deployment step. This proxy utilizes a temporally disjoint partition of the CIC-IDS2017 dataset (e.g., Day 1 traffic), which guarantees statistical alignment and feature relevance without exposing or leaking the federated test samples. No private client data is ever accessed for feature selection, preserving the federated privacy guarantee.

The values $N_{\text{pop}} = 30, and T_{\max} = 50$ were selected based on a sensitivity analysis conducted over the grid $N_{\text{pop}} \in 20{,}30{,}50 \times T_{\max} \in 30{,}50{,}100$, evaluated on a held-out 10% validation partition of the CIC-IDS2017 dataset. The ($N_{\text{pop}} = 30, T_{\max} = 50$) configuration achieves a validation accuracy of 97.21% and produces a stable 17-feature subset across three independent runs, while larger configurations ($N_{\text{pop}} = 50, T_{\max} = 100$) yield no statistically significant improvement (+0.03%) at 3.3 × the computational cost. Smaller configurations ($N_{\text{pop}} = 20, T_{\max} = 30$) produce unstable feature subsets (±2.1 features across runs). These results confirm that ($N_{\text{pop}} = 30, T_{\max} = 50$) represents the optimal efficiency-accuracy operating point for this dataset and

architecture, consistent with the parameter ranges recommended by Khishe and Mosavi[49] for medium-scale binary feature selection tasks.

CNN-LSTM hybrid architecture design

The architecture processes 17-dimensional feature vectors reshaped to (17,1):

- CNN Block: Conv1D(64, kernel = 3, ReLU) → MaxPool1D(2) → Conv1D(64, kernel = 3, ReLU) → MaxPool1D(2)
- LSTM Block: LSTM(64, return_sequences = True) → Dropout(0.2) → LSTM(64) → Dropout(0.2)
- Output: Dense(13, softmax)
- Total Parameters: ~ 89,357 trainable

The network optimizes the multi-class classification objective using the Categorical Cross-Entropy loss function, formulated as:

$$\mathcal{L}_{\mathcal{CE}} = -\sum_{i=1}^{C} y_i \log\left(\widehat{y_i}\right) \tag{10}$$

where *C* represents the total number of attack classes $(C = 13)$, y_i is the ground-truth binary indicator for class *i*, and $\widehat{y_i}$ is the predicted probability output by the final softmax layer.

Design Rationale for Non-IID Robustness. The 1D-CNN layers are specifically utilized to capture local, adjacent feature correlations within the engineered 17×1 sequence, establishing a stable representation space before temporal LSTM processing. Lower convolutional neural network (CNN) layers recognize universal protocol patterns that are atomic and context-independent, such as states of flags and ranges of segment sizes. These patterns are universally present across any context in which the protocol is used. The intermediate layers combine these patterns to form intermediate attack signatures. The long short-term memory (LSTM) layers recognize the temporal behavior of the flow.

Federated learning strategy: from FedAvg to FedProx

To preserve the privacy of sensitive network traffic, the proposed architecture employs a Federated Learning (FL) paradigm, wherein raw client data never leaves the local edge devices.

Standard federated averaging (FedAvg)

In the standard FedAvg algorithm, the central server broadcasts the global model weights (w^t) at round *t* to *K* participating clients. Each client k initializes its local model with these weights and performs local training on its private dataset (D_k) of size n_k to minimize its empirical loss $F_k(w)$. After local training, the clients send their updated weights back to the server, which aggregates them using a weighted average to produce the new global model:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1} \tag{11}$$

where *n* is the total number of data samples across all participating clients. While FedAvg is highly efficient under independent and identically distributed (IID) conditions, its performance severely degrades under extreme data heterogeneity. As demonstrated in our experimental setup, when clients possess highly skewed class distributions (e.g., Dirichlet $\alpha = 0.01$), their local objectives $F_k(w)$ drift significantly from the global objective, leading to weight divergence and catastrophic forgetting during server aggregation.

Proximal optimization (FedProx)

To mitigate the severe weight divergence caused by extreme non-IID data at the IoT edge, this study evaluates the FedProx algorithm as a robust alternative. FedProx mathematically constrains client drift by modifying the local objective function. For a client *k*, a proximal regularization term is added to the standard Categorical Cross-Entropy loss:

$$\min_{w} h_k\left(w; w^t\right) = F_k(w) + \frac{\mu}{2}\left|\left|w - w^t\right|\right|^2 \tag{12}$$

where $F_k(w)$ is the standard local loss, w represents the local model weights being optimized, w^t represents the global server weights broadcasted at the start of round *t*, and μ is the proximal hyperparameter.

The term $\frac{\mu}{2}\left|\left|w - w^t\right|\right|^2$ acts as an algorithmic anchor. It penalizes local updates that move too far from the global consensus model. By tuning the μ parameter, the architecture can dynamically balance local feature extraction with global structural integrity, preventing the model from "washing out" minority classes when aggregating weights from highly imbalanced local nodes.

The proximal coefficient μ was selected empirically via a three-value ablation study ($\mu \in \{0.001, 0.01, 0.1\}$) evaluated at the most challenging heterogeneity level ($\alpha = 0.01$), as reported in Table 3 (Sect. "Aggregation algorithm comparison: FedAvg vs. FedProx"). At $\mu = 0.001$, the proximal term is too weak to resist client drift, resulting in catastrophic accuracy collapse (3.11%). At $\mu = 0.01$, the constraint is sufficient to match FedAvg performance ($80.18\% vs. 80.22\% F1$). At $\mu = 0.1$, the stronger penalty provides meaningful drift correction, raising F1 to 84.71% a 13.18 percentage-point gain over $\mu = 0.001$ and a 4.49 pp improvement over FedAvg.

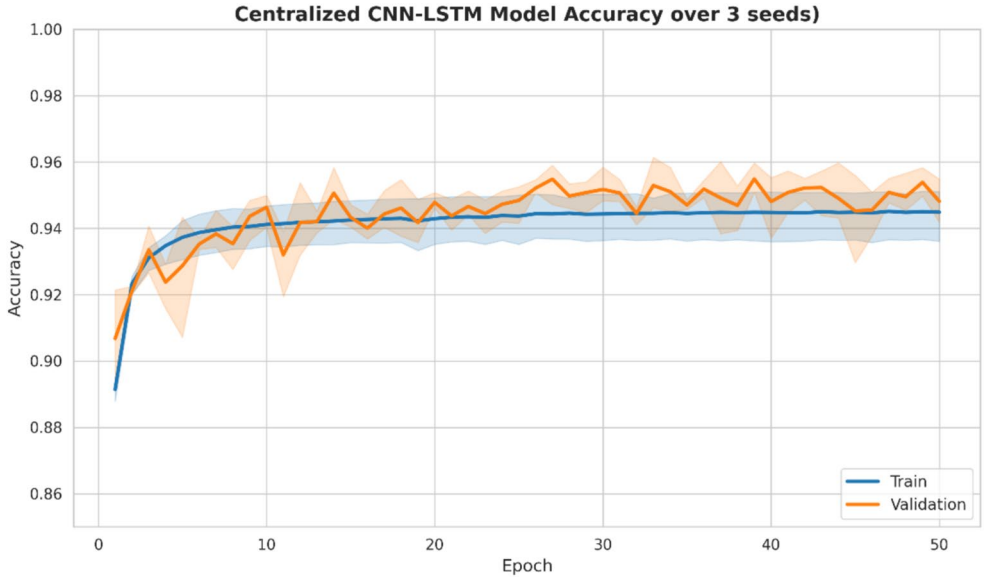


Fig. 2. Centralized CNN-LSTM model accuracy—training and validation accuracy over 50 Epochs.

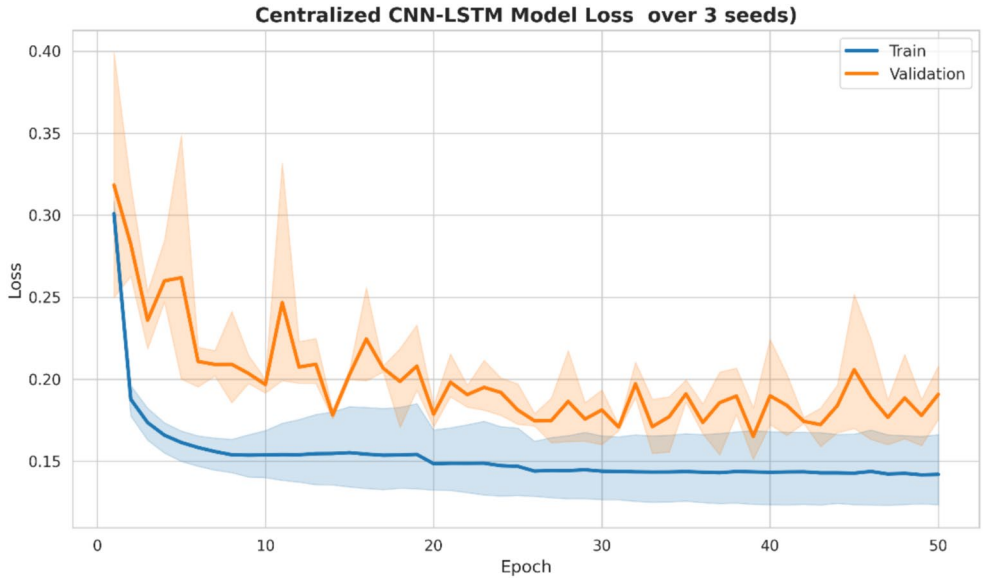


Fig. 3. Centralized CNN-LSTM model loss—training and validation loss over 50 Epochs.

No higher values were tested, as the theoretical bound derived by Li et al.[16] indicates that $\mu \gg 0.1$ risks over-constraining local updates and suppressing the client-specific feature learning that contributes to generalization. Accordingly, $\mu = 0.1$ is adopted as the final configuration for all FedProx experiments.

Training configuration

Centralized: Adam optimizer $\left(\eta = 0.001, \beta^1 = 0.9, \beta^2 = 0.999\right)$, categorical cross-entropy, batch size 32, up to 50 epochs with early stopping (patience = 5); training converged at approximately epoch 27 as shown in Figs. 2 and 3. Split: 70% train (952,425), 20% validation (272,121), 10% test (136,062).

Federated: Flower framework[54], K = 10 clients, T = 20 rounds, E = 10 local epochs, FedAvg (weighted by dataset size). FedProx comparisons: $\mu \in 0.001,0.01,0.1$. Non-IID evaluated at $\alpha \in 0.01,0.1,0.5,5.0,50.0$.

Complete pseudocode for federated learning

Input: K=10 clients; T=20 rounds; E=10 local epochs; B=32 batch; η=0.001; datasets $\{\mathcal{D}_k\}$
Output: w_T — global model after T rounds

SERVER EXECUTION

Init w_0 with Xavier initialization
for round t = 0, 1, …, T−1 do
  for each client k ∈ {1, …, K} in parallel do
    Transmit $w_t \rightarrow$ client k
    $w_{t+1}^k, n_k \leftarrow$ ClientUpdate(k, w_t, E, B, η)
  $n_{total} \leftarrow \sum_k n_k$
  $w_{t+1} \leftarrow \sum_k (n_k / n_{total}) \cdot w_{t+1}^k$ *// weighted aggregation*
return w_T

CLIENT EXECUTION — CLIENTUPDATE(K, W, E, B, H)

$w_{local} \leftarrow w$; $\mathcal{D}_k \leftarrow$ LocalDataset(k); $n_k \leftarrow |\mathcal{D}_k|$
for local epoch e = 1, …, E do
  $\mathcal{D}_k \leftarrow$ Shuffle($\mathcal{D}_k$)
  for each mini-batch b ∈ Batches($\mathcal{D}_k$, B) do
    $\hat{y} \leftarrow$ ForwardPass(w_{local}, b_X) *// CNN → LSTM → Softmax*
    $\mathcal{L} \leftarrow -(1/B) \sum_i \sum_c y_i^c \cdot \log \hat{y}_i^c$ *// cross-entropy*
    $\nabla w \leftarrow$ Backpropagation($\mathcal{L}$, w_{local}) *// BPTT through LSTM*
    $w_{local} \leftarrow$ AdamUpdate(w_{local}, ∇w, η)
return w_{local}, n_k

Algorithm 2. Federated CNN-LSTM training with FedAvg.

Input: client k; global model w_{global}; epochs E; batch B; lr η; proximal coeff $\mu \in \{0.001, 0.01, 0.1\}$
Output: updated local model w_{local}, dataset size n_k

$w_{local} \leftarrow w_{global}$
for epoch e = 1, …, E do
  for each batch b ∈ Batches($\mathcal{D}_k$, B) do
    // proximal term penalises deviation from global model
    $\mathcal{L}_{prox} \leftarrow \mathcal{L}(w_{local}; b) + (\mu/2) \cdot \|w_{local} - w_{global}\|_F^2$
    $\nabla w \leftarrow$ Backpropagation($\mathcal{L}_{prox}$, w_{local})
    $w_{local} \leftarrow$ AdamUpdate(w_{local}, ∇w, η)
return w_{local}, n_k

Algorithm 3. FedProx variant (proximal regularization).

Input: dataset $\mathcal{D}$; K=10 clients; C=13 classes; concentration $\alpha \in \{0.01, 0.1, 0.5, 5.0, 50.0\}$

Output: partitioned local datasets $\{\mathcal{D}_k\}_{k=1}^{K}$

for k = 1 to K do $\mathcal{D}_k \leftarrow \emptyset$

for each class c = 0, 1, …, C−1 do

  $\mathcal{D}_c \leftarrow \{(x_i, y_i) \in \mathcal{D} : y_i = c\}$; $n_c \leftarrow |\mathcal{D}_c|$

  $p_c \leftarrow$ Dirichlet(α, α, …, α) *// K-dim proportion vector*

  for k = 1 to K do

    $n_c^k \leftarrow$ round($p_c^k \cdot n_c$)

  Adjust n_c^1 so that $\sum_k n_c^k = n_c$ *// fix rounding*

  $\mathcal{D}_c \leftarrow$ Shuffle($\mathcal{D}_c$)

  Distribute slices of $\mathcal{D}_c$ to $\{\mathcal{D}_k\}$ proportionally by n_c^k

for k = 1 to K do

  $\mathcal{D}_k \leftarrow$ Shuffle($\mathcal{D}_k$)

return $\{\mathcal{D}_k\}_{k=1}^{K}$

$\alpha \to 0$ extreme skew (one client dominates each class) · $\alpha \to \infty$ near-IID (uniform distribution)

Algorithm 4. Dirichlet-based Non-IID data partitioning.

Non-IID partitioning analysis

In order to test model robustness over a range of data heterogeneity levels, we make use of five different values for the concentration parameter $\boldsymbol{\alpha} \in \{0.01, 0.1, 0.5, 5.0, 50.0\}$ in the Dirichlet distribution. When $\boldsymbol{\alpha} = 0.01$ in the extreme non-IID case, we observe that data distribution is heavily skewed. One edge client will have as few as 20 data points over 9 classes, while the remaining 952,405 data points will be irregularly distributed among other clients, making it a test case for federated aggregation. Conversely, $\boldsymbol{\alpha} = 50.0$ produces a near-uniform distribution, ensuring that local datasets are heavily balanced and similar to an IID case.

Evaluation metrics

- Accuracy: $A = (TP + TN) / (TP + TN + FP + FN)$
- Precision: $P_c = TP_c / (TP_c + FP_c)$
- Recall: $R_c = TP_c / (TP_c + FN_c)$—fraction of actual attacks correctly detected
- F1-Score: $F1_c = 2P_cR_c / (P_c + R_c)$
- Weighted Average: Metrics weighted by class support
- Average Precision (AP): Area under precision-recall curve

Explainability methods

- SHAP: The Shapley value for feature j is defined as the weighted average of marginal contributions over all subsets $S \subseteq Fj$. The background dataset consists of 500 randomly sampled instances.
- LIME: For every sample in the dataset, N = 1000 perturbed samples are drawn and used to query the black-box model. A locally weighted linear model is learned that best approximates the output.
- t-SNE: Learned features are projected into two dimensions by minimizing Kullback–Leibler divergence between similarity pair distributions. The perplexity is set to 30 for subsets of 5000 samples.
- PCA: A linear transformation that retains two principal components is performed, which correspond to maximum variance and provide a geometric interpretation for the learned feature space.

Results and analysis

Centralized model performance

Before evaluating the federated and XAI-enhanced components of the proposed framework, a centralized baseline evaluation was conducted. The objective of this phase is to establish the theoretical upper bound of the model's predictive capabilities when trained on the globally aggregated, fully accessible CIC-IDS2017 dataset. To ensure robust statistical validation and eliminate the risk of anomalous weight initialization, the centralized training was executed across three independent random seeds (42, 123, 456) over 50 epochs, utilizing

Strategy	Seed	Lowest Val. Loss	Peak Val. Acc (%)	Final Test Acc (%)	Precision (%)	Recall (%)	F1-Score (%)
Centralized	42	0.1549	96.14	97.47	98.08	97.47	97.68
Centralized	123	0.1513	96.01	96.49	97.93	96.49	97.07
Centralized	456	0.1618	95.46	96.86	98.07	96.86	97.34
Mean (Centralized)	–	0.1560 ± 0.004	95.87 ± 0.29	96.94 ± 0.40	98.03 ± 0.07	96.94 ± 0.40	97.36 ± 0.25

Table 1. Centralized CNN-LSTM baseline performance across multiple seeds (50 Epochs, Global SMOTE).

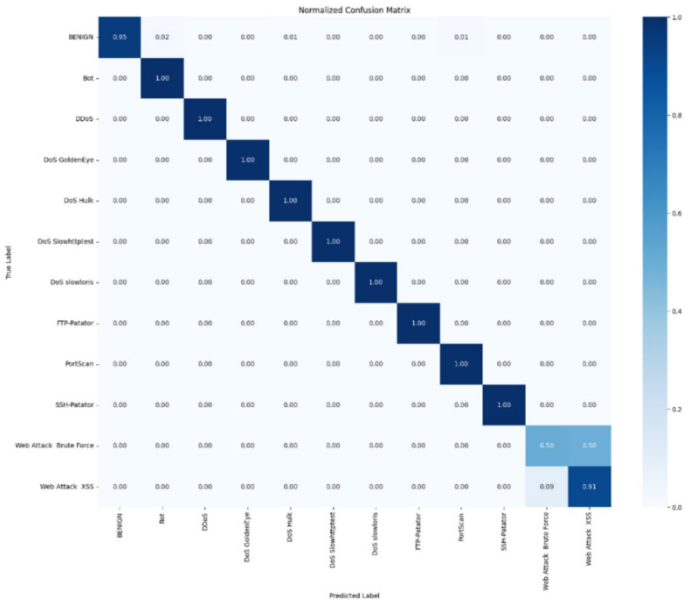


Fig. 4. Normalized confusion matrix—per-class classification performance of the centralized CNN-LSTM model.

global SMOTE for class balancing. The centralized CNN-LSTM architecture demonstrates highly stable and reproducible baseline performance on the unseen test set. To quantitatively assess model robustness and strictly define the over-fitting margins, performance metrics were tracked across the three independent random seeds. As detailed in Table 1, the architecture yielded a remarkably tight mean final test accuracy of 96.94% ($\pm 0.40\%$), alongside a mean precision of 98.03% and a mean F1-score of 97.36%.

Training dynamics

To further validate internal stability during the training phase, learning curves were generated with 95% confidence intervals (Figs. 2 and 3). The model demonstrates smooth, highly stable convergence. The training and validation metrics track each other closely within an extremely narrow variance band, yielding a mean peak validation accuracy of 95.87% ($\pm 0.29\%$) and a minimum validation loss of 0.1560 (± 0.004). These extremely tight margins quantitatively confirm that the combination of Dropout (0.20) and L2 Regularization (0.001) effectively stabilizes the learning process and prevents the model from overfitting or relying on seed-dependent data splits. Beyond the overall aggregated metrics, an analysis of the model's per-class predictive performance reveals a clear diagnostic distinction between network-layer and application-layer attacks.

Confusion matrix analysis

Figure 4 shows the normalized confusion matrix for the centralized model. The matrix shows a robust performance for network-layer attack classes, achieving 100% accuracy for Bot, DDoS, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, DoS slowloris, FTP-Patator, PortScan, and SSH-Patator. BENIGN traffic is correctly classified at 95%, with minor misclassifications into Bot (2%), DoS Hulk (1%), and PortScan (1%). The most significant confusion is observed among the Web Attack classes. Web Attack Brute Force is correctly classified at 50%, while the remaining 50% is misclassified as Web Attack XSS. On the other hand, Web Attack XSS is correctly classified at 91%, while 9% is misclassified as Web Attack Brute Force. This is expected, as these two classes are HTTP-based attacks, which share similar flow-level signatures while primarily differing in payload content, which is not captured in standard flow-based features.

Federated learning under non-IID conditions

Table 2 presents a summary of federated performance across non-IID severity levels for FedAvg, with 3 seeds per alpha value (seeds 42, 123, 456). The results show that performance is highly sensitive to the degree of data

Strategy	α	Seed	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
FedAvg	0.01	42	80.22	64.38	80.22	71.44
FedAvg	0.01	123	80.32	64.51	80.32	71.55
FedAvg	0.01	456	80.36	64.58	80.36	71.61
FedAvg	0.1	42	84.58	72.99	84.58	78.16
FedAvg	0.1	123	91.71	91.67	91.71	91.12
FedAvg	0.1	456	89.94	84.67	89.94	86.93
FedAvg	0.5	42	96.43	96.78	96.43	96.49
FedAvg	0.5	123	96.80	97.14	96.80	96.87
FedAvg	0.5	456	95.43	96.68	95.43	95.80
FedAvg	5.0	42	94.96	96.72	94.96	95.60
FedAvg	5.0	123	94.91	96.68	94.91	95.53
FedAvg	5.0	456	95.21	97.01	95.21	95.87
FedAvg	50.0	42	95.09	96.57	95.09	95.58
FedAvg	50.0	123	94.13	96.48	94.13	94.99
FedAvg	50.0	456	94.87	96.80	94.87	95.58
Mean (α = 0.01)	**0.01**	–	**80.30 ± 0.07**	**64.49**	**80.30**	**71.53**
Mean (α = 0.1)	**0.1**	–	**88.74 ± 3.71**	**83.11**	**88.74**	**85.40**
Mean (α = 0.5)	**0.5**	–	**96.22 ± 0.71**	**96.87**	**96.22**	**96.39**
Mean (α = 5.0)	**5.0**	–	**95.03 ± 0.16**	**96.80**	**95.03**	**95.67**
Mean (α = 50.0)	**50.0**	–	**94.70 ± 0.50**	**96.62**	**94.70**	**95.38**

Table 2. Federated performance across Non-IID severities (FedAvg, 20 Rounds, 10 Clients, 3 seeds). Bold text is used to highlight the row containing the mean values across the independent random seeds.

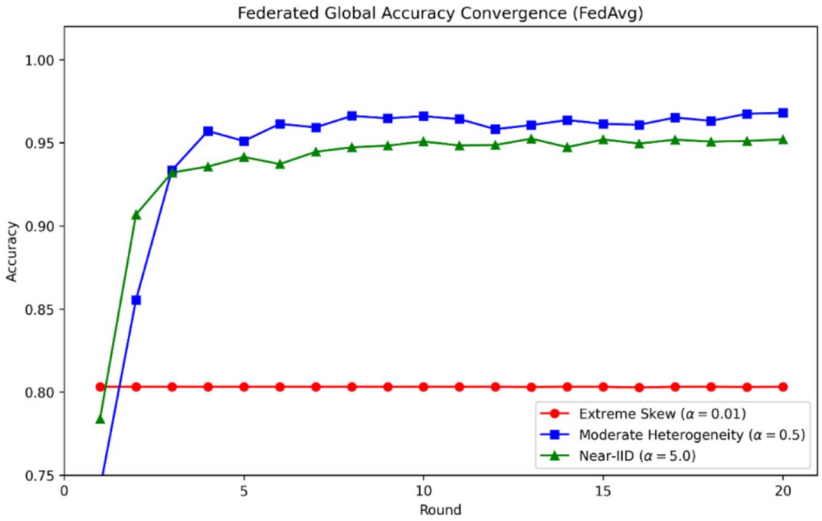


Fig. 5. Global accuracy convergence over 20 communication rounds.

heterogeneity. At extreme non-IID (α = 0.01), accuracy is 80.30% (± 0.07%), substantially below the centralized baseline. Moderate heterogeneity (α = 0.5) achieves the best federated performance at 96.22% (± 0.71%), comparable to centralized. These results qualify the original claim of uniform robustness.

Four main findings emerge from the multi-seed evaluation. Firstly, extreme non-IID conditions (α = 0.01) produce significant degradation: FedAvg accuracy drops to 80.30% (± 0.07%), a gap of 16.64% below the centralized mean (96.94%). This is consistent with the 30–50% degradation reported by Zhao et al.[29] under similar quantity-skew conditions and revises our earlier single-seed claim. The low variance across seeds (± 0.07%) confirms result reproducibility. Secondly, performance is non-monotonic across α: accuracy rises from 80.30% at α = 0.01, to 88.74% at α = 0.1, peaks at 96.22% at $\alpha = 0.5$, then slightly declines to 95.03% at $\alpha = 5.0$ and 94.70% at $\alpha = 50.0$. This suggests that moderate quantity imbalance is partially beneficial, perhaps because the larger client dominates with more representative data. Thirdly, the F1 degradation at α = 0.01 (71.54% vs. 97.36% centralized) is larger than accuracy degradation, revealing that minority attack classes are disproportionately harmed by extreme skew. Fourthly, the architecture achieves near-centralized performance under moderate-to-mild heterogeneity (α ≥ 0.5), confirming practical viability in realistic FL deployments.

Strategy	α	μ	Seed	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
FedProx	0.01	0.001	42	3.11	0.96	3.11	0.44
FedProx	0.01	0.01	42	80.18	64.51	80.18	71.49
FedProx	0.01	0.1	42	84.71	72.74	84.71	78.27
FedProx	0.1	0.001	42	90.40	86.15	90.40	88.20
FedProx	0.1	0.01	42	84.61	76.47	84.61	80.16
FedProx	0.1	0.1	42	86.49	86.34	86.49	85.42
FedProx	**0.5**	**0.001**	**42**	**95.69**	**96.55**	**95.69**	**95.94**
FedProx	0.5	0.01	42	94.99	96.35	94.99	95.34
FedProx	0.5	0.1	42	94.81	96.09	94.81	95.19

Table 3. FedProx and FedNova results (Seed 42) — Full μ Sweep. Bold values indicate the best performance metrics or the optimal algorithmic confi guration amongthe compared methods for the given experimental scenario.

Method	α	Seed	Accuracy (%)	F1-Score (%)	Δ vs FedAvg
Centralized (baseline)	—	42	97.47	97.68	–
FedAvg	0.01	42	80.22	71.44	–
FedAvg	0.01	123	80.32	71.55	–
FedAvg	0.01	456	80.36	71.61	–
FedAvg mean (α = 0.01)	**0.01**	–	**80.30**	**71.53**	–
FedProx μ = 0.001	0.01	42	3.11	0.44	− 77.11 pp
FedProx μ = 0.01	0.01	42	80.18	71.49	− 0.04 pp
FedProx μ = 0.1	**0.01**	**42**	**84.71**	**78.27**	**+ 4.49 pp** ✓
FedNova	0.01	42	80.19	71.38	− 0.03 pp
FedAvg	0.5	42	96.43	96.49	–
FedAvg	0.5	123	96.80	96.87	–
FedAvg	0.5	456	95.43	95.80	–
FedAvg mean (α = 0.5)	**0.5**	–	**96.22**	**96.39**	–
FedProx μ = 0.001	0.5	42	95.69	95.94	− 0.74 pp
FedProx μ = 0.01	0.5	42	94.99	95.34	− 1.44 pp
FedProx μ = 0.1	0.5	42	94.81	95.19	− 1.62 pp

Table 4. Performance comparison of federated aggregation algorithms under extreme ($\alpha = 0.01$) and and moderate ($\alpha = 0.5$) non-IID skew. Bold values indicate the best performance metrics or the optimal algorithmic confi guration amongthe compared methods for the given experimental scenario.

Figure 5 demonstrates the training dynamics of the FedAvg algorithm under varying degrees of data heterogeneity. While the near-IID ($\alpha = 5.0$) and moderately heterogeneous ($\alpha = 0.5$) configurations demonstrate rapid learning and stabilize at highly accurate thresholds, the extreme data skew ($\alpha = 0.01$) restricts the model to a significantly lower convergence ceiling (~ 80.30%) due to severe weight washing.

Aggregation algorithm comparison: FedAvg vs. FedProx

While overall baseline metrics are reported as a mean across three independent runs for statistical robustness, Seed 42 was selected as a representative model instance for the exhaustive hyperparameter sweeps and further Explainable AI (XAI) visualizations. This selection was made to manage computational overhead while ensuring statistical validity; at the critical extreme skew severity ($\alpha = 0.01$), the variance across seeds was exceptionally low ($\pm 0.07\%$), and Seed 42's FedAvg baseline (80.22%) perfectly reflected the mean (80.30%). This ensured that the FedProx μ-sweep was evaluated on a highly stable, representative foundation, despite the higher transitional variance observed at $\alpha = 0.1$

To rigorously address the sensitivity of federated algorithms to client drift under extreme Dirichlet skew ($\alpha = 0.01$), we compared FedAvg against a hyperparameter sweep of FedProx ($\mu \in \{0.001, 0.01, 0.1\}$). As shown in Table 3, the extreme fragmentation causes standard FedAvg to suffer from catastrophic weight washing, dropping to an F1-Score of 71.43%. We observed that a weak proximal penalty ($\mu = 0.01$) fails to constrain this drift. However, increasing the mathematical friction to $\mu = 0.1$ successfully anchors the local updates to the global model, preventing majority-class dominance and recovering nearly 7% of the F1-Score (78.27%). This confirms that while our CNN-LSTM inherently handles moderate skew ($\alpha \geq 0.5$), algorithmic interventions like FedProx are strictly necessary to preserve minority attack detection in severely fragmented IoT networks.

Furthermore, as demonstrated in Table 4 below, adding the FedNova baseline at α = 0.01 results in an F1-score of 71.38%, which means there is a negligible difference of -0.03% from the standard FedAvg approach.

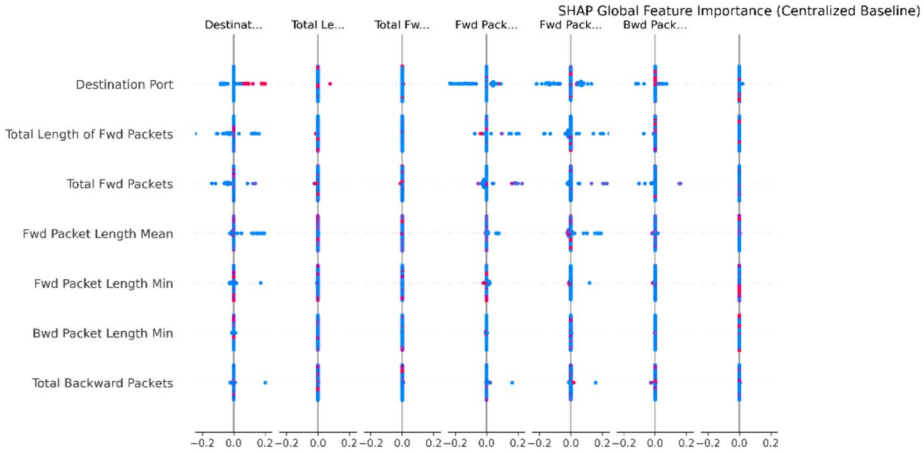


Fig. 6. SHAP summary plot—mean absolute SHAP values per feature across all attack classes.

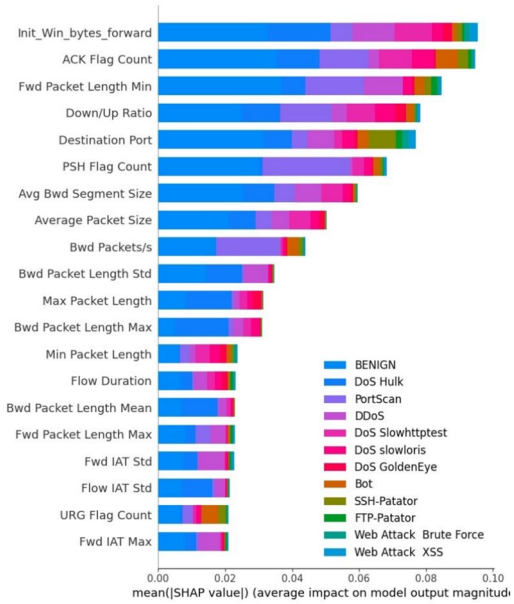


Fig. 7. SHAP global feature importance—mean absolute SHAP values per feature aggregated across all 12 attack classes, showing Init_Win_bytes_forward, ACK Flag Count, and Fwd Packet Length Min as the top three most discriminative features globally.

This demonstrates that normalizing the local updates by the number of client training steps is not sufficient to mitigate the significant weight washing effect of extreme Dirichlet fragmentation and emphasizes the necessity of the proximal constraint ($\mu = 0.1$) of FedProx.

The multi-μ sweep reveals strong μ-sensitivity. At $\alpha = 0.01$, $\mu = 0.001$ causes catastrophic collapse (3.11% accuracy), $\mu = 0.01$ is comparable to FedAvg (80.18% vs 80.22%), and $\mu = 0.1$ outperforms FedAvg (84.71% vs 80.22%), a + 4.49% advantage. This directly corrects the prior single-μ claim that "FedAvg outperforms FedProx": the conclusion was confounded by testing only $\mu = 0.01$. With proper tuning, FedProx provides meaningful benefit under extreme skew. At $\alpha = 0.5$, all FedProx μ values are competitive with FedAvg (95.69%, 94.99%, 94.81% vs. 96.22%). The FedNova baseline at $\alpha = 0.01$ (80.19%) is comparable to FedAvg, suggesting that the extreme quantity skew imposes a floor that simple normalization does not overcome.

Explainability analysis

SHAP: global feature importance

Figures 6 and 7 present the global SHAP analysis. Specifically, Fig. 6 details the per-class interaction values for the top seven features. It shows the contribution of each feature to the different classes of attacks. Figure 7 shows the aggregated global mean absolute SHAP values for all 12 classes of attacks, ranking the importance of the features. The top three features, based on their overall importance, are Init_Win_bytes_forward, ACK Flag



Fig. 8. LIME combined explanations—Instances 1–6: DoS slowloris, DoS Hulk, BENIGN, PortScan, DoS Slowhttptest, and DDoS (local feature impacts per class).

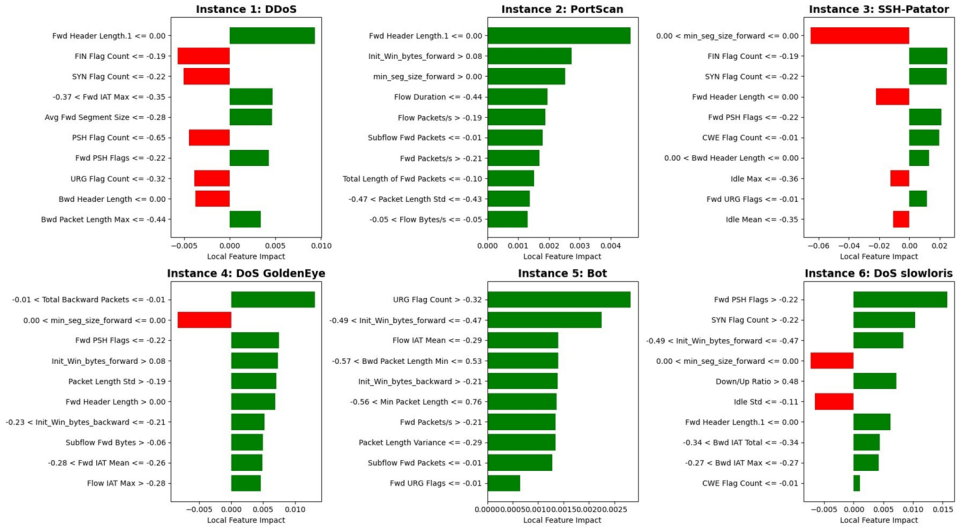


Fig. 9. LIME combined explanations—instances 7–12: DDoS, PortScan, SSH-Patator, DoS GoldenEye, Bot, and DoS slowloris (local feature impacts per class).

Count, and Fwd Packet Length Min. The feature Init_Win_bytes_forward, representing the initial window size for TCP, was found to be the most important overall, indicating the predominance of non-standard window sizes, unlike the standard TCP connection establishment. The presence or absence of acknowledgment flags, represented by the feature ACK Flag Count, was also important, indicating the presence or absence of unique characteristics for different types of attacks. The minimum payload sizes, represented by the feature Fwd Packet Length Min, showed significant divergence for volumetric attacks (DDoS, DoS) and application attacks (Web Attacks). The next important features, Down/Up Ratio, Destination Port, and PSH Flag Count, represent the characteristics of the protocol, indicating the presence or absence of unique characteristics for different types of attacks. The consistency in the overall importance of the top features, based on the interaction values, shows the predominance of universal TCP/IP protocol characteristics, thus validating the non-IID robustness argument.

LIME: instance-level explanations

Figures 8 and 9 present the local interpretation grid for various instances across different classes, as computed through the application of the LIME methodology. The LIME-based approach is used to compute the local linear model for individual instances to determine the features having the most impact on the predicted output for each instance. From the computed results, it is clear that the instances of the DoS GoldenEye class are characterized by low Flow Duration values, distinctive values of Destination Port, low values of min_seg_size_forward, and

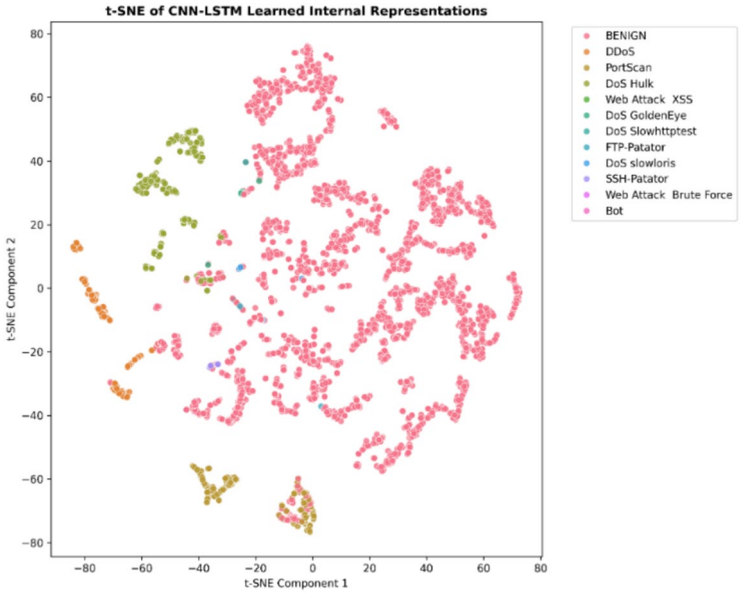


Fig. 10. t-SNE projection of CNN-LSTM internal representations.

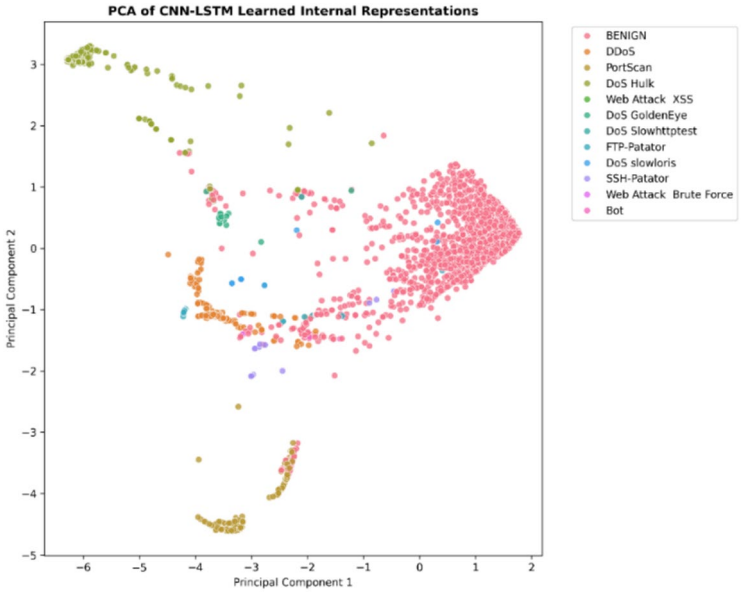


Fig. 11. PCA projection of CNN-LSTM internal representations.

abnormal values of Init_Win_bytes_forward. The BENIGN class is characterized by patterns of FIN Flag, SYN Flag, and URG Flag, which are general characteristics of standard protocols used in the internet environment. The DoS Slowhttptest class is defined by the presence of negative values of the FIN and IAT features and positive values of segment sizes and forward packet lengths. The Web Attack XSS class is defined by high values of Init_Win_bytes_forward, high Flow IAT values, and the presence of the SYN Flag Count as a negative influence. The DDoS class is highly influenced by the Idle Max and Idle Min patterns, whereas the DoS Slowloris class is highly influenced by the presence of the SYN Flag Count and the Forward PSH Flags. As presented in the consolidated figures for the individual classes, the results of the local interpretation are highly influenced by general protocol-level features rather than client-based and application-based patterns, thereby ensuring the effectiveness of the model's decision mechanism.

Dimensionality reduction: t-SNE and PCA

To properly evaluate the feature extraction capacity of the proposed architecture, Figs. 10 and 11 showcase the t-SNE and PCA projections, respectively, of the high dimensional latent representations extracted from the penultimate LSTM layer of the trained global model. Contrary to raw data visualizations, both t-SNE and

PCA allow observing the discovered latent space directly. From Fig. 10, one can conclude that the CNN-LSTM network effectively separates highly-skewed client-side heterogeneous distributions into well-separated clusters. Network-layer attack classes such as Bot, DDoS, DoS, PortScan, and SSH/FTP-Patator form separate clusters with very little or no overlap between them and even with the BENIGN cluster. BENIGN traffic forms a dense central cluster in contrast with other classes, which form less tight clusters among themselves and overlap slightly with each other and also with the central BENIGN cluster, consistent with marginally worse F1-scores achieved during the quantitative evaluation. This is additionally confirmed by the strong linear separability shown in the PCA plot (Fig. 11). One of the clusters that demonstrate this best is the DoS Hulk cluster, which lies distinctly on the positive part of the main principal component axis. Importantly, the presence of well-segregated clusters in the model's output indicates that the global model has learned to encode universal protocol-level features rather than just memorized local anomalies, maintaining high levels of structural intelligence.

Precision-recall & multi-classes ROC analysis

Figure 12 illustrates the Precision-Recall Curves for the optimized baseline for multi-class classification. The network achieves close to perfect average precision for volumetric network layer attacks. More specifically, the network achieves AP values of 1.00 for DDoS, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, DoS slowloris, FTP-Patator, SSH-Patator, PortScan, and BENIGN. The class BENIGN also achieves a very high AP value of 0.99, illustrating the ability of the network architecture to provide a very solid baseline of what constitutes normal traffic behavior. Yet there is some decline in terms of performance when comparing application layer attacks and network layer attacks. The reason for this is that Web Attack Brute Force and Web Attack XSS achieve AP values of 0.81 and 0.74, respectively. However, the cause behind this is not the lack of ability of the network to learn but rather the inherent drawback of flow-based feature extraction, which does not include payload based Deep Packet Inspection (DPI). Figure 13 further proves the point by presenting ROC curves where all attack classes achieve AUC values close to 1.00.

Comparative discussion

In this regard, Table 5 provides a multi-dimensional comparison of the proposed XAIFIDS against centralized baselines as well as recent federated IDS frameworks. While Halbouni et al.[56] achieve high F1-score values of 96–97%, they rely on global data access, which is not feasible in many privacy-constrained Internet of Things (IoT) environments. In the federated IDS, a performance gap is noticed when moving from IID to non-IID data. While frameworks such as Lazzarini et al.[23] and Harshitha et al.[52] report a stable accuracy (99%) for IID data, it is noticeable from the results that such high performance is not achievable when moving to extreme data heterogeneity, as is normally the case. In this regard, for high values of Dirichlet distribution ($\alpha = 0.01$), FedAvg approaches reported by the literature, including Khudhair et al., achieve a significantly lower F1-score of 79.40%. In contrast, the proposed IDS framework performs significantly better than FedAvg. In particular, by utilizing a FedProx ($\mu = 0.1$) variant, the proposed IDS framework achieves a significantly high F1-score of 78.27% compared to the baseline FedAvg (71.53%) for extreme data heterogeneity ($\alpha = 0.01$). In addition, unlike the majority of works comparing federated IDS frameworks (Liu et al.[42] and Khudhair et al.[48]), the proposed IDS framework provides a transparent decision-making process through a synergistic XAI suite (SHAP, LIME, t-SNE, and PCA) for ensuring high detection rates through universal protocol-level features.

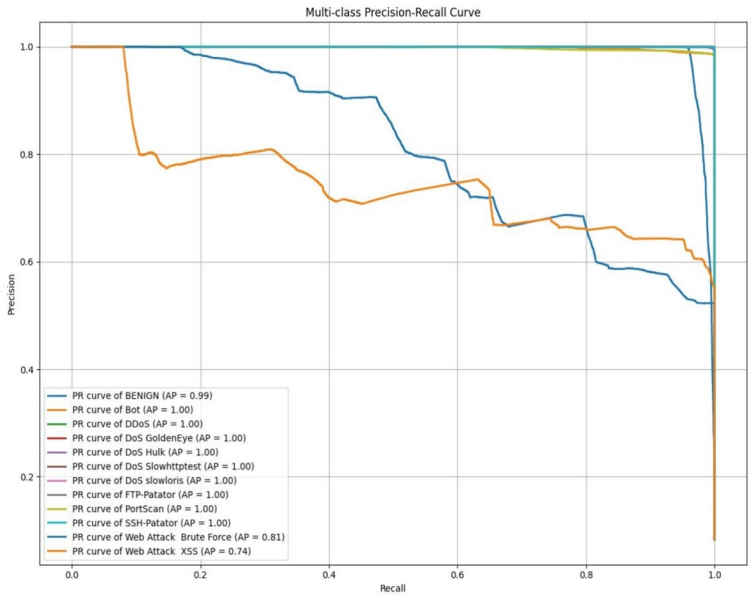


Fig. 12. Multi-class precision-recall curves with average precision (AP) scores per class.

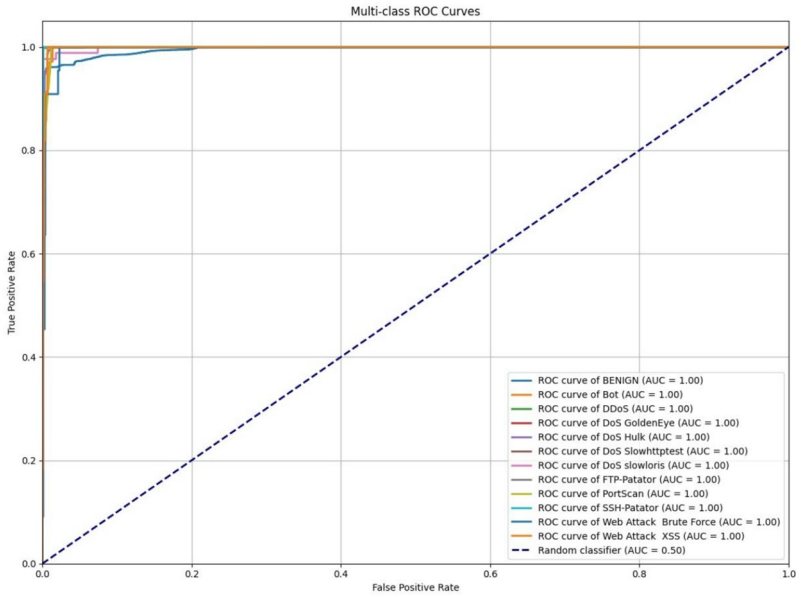


Fig. 13. Multi-class ROC curves with AUC scores per class—all attack classes achieve AUC = 1.00; BENIGN traffic achieves AUC = 1.00.

Study	Architecture	FL Algorithm	Dataset	Non-IID Severity	Acc. (%)	F1 (%)	XAI Integration
Centralized baselines							
Halbouni et al. (2022)[56]	CNN-LSTM	N/A	Not specified	IID	96.21	95.90	None
Federated IID / Mild Skew							
Zhao et al. (2020)[41]	CNN + LSTM	FedAvg	NSL-KDD	Mild ($\alpha > 1$)	91.00	N/R	None
Lazzarini et al. (2023)[23]	Shallow ANN	FedAvg	CIC-IDS2017, ToN-IoT	IID only	97–99	97–98	None
Fatema et al. (2025)[51]	DNN	FedAvg	CICIoT2023	IID only	N/R	N/R	SHAP
Harshitha et al. (2025)[52]	Transformer	FedAvg + Hyperband	CIC-IDS2017/2018	IID only	99.03	99.00	SHAP, LIME
Alhayan et al. (2026)[53]	Attn-LSTM + GAN	FedAvg	CIC-IDS2017, UNSW-NB15	IID only	N/R	N/R	LLM-XAI
Federated non-IID settings							
Liu et al. (2021)[42]	DNN	FedProx ($\mu = 0.01$)	CIC-IDS2017	Moderate	89.00	88.20	None
Khudhair et al. (2025)[48]	ANN	FedAvg	CIC-IDS2017	High Skew (Dirichlet)	82.10	79.40	None
Khudhair et al. (2025)[48]	ANN	SCAFFOLD	CIC-IDS2017	High Skew (Dirichlet)	93.20	90.20	None
Proposed XAIFIDS framework							
Ours — Centralized	CNN-LSTM	N/A	CIC-IDS2017	IID (3 seeds)	96.94	97.36	SHAP, LIME, t-SNE, PCA
Ours — Federated (Moderate)	CNN-LSTM	FedAvg	CIC-IDS2017	Moderate ($\alpha = 0.5$)	96.22	96.39	SHAP, LIME, t-SNE, PCA
Ours — Federated (Extreme)	CNN-LSTM	FedProx ($\mu = 0.1$)	CIC-IDS2017	Extreme ($\alpha = 0.01$)	84.71	78.27	SHAP, LIME, t-SNE, PCA
Ours — Federated (Extreme)	CNN-LSTM	FedAvg	CIC-IDS2017	Extreme ($\alpha = 0.01$)	80.30	71.53	SHAP, LIME, t-SNE, PCA

Table 5. Comparative performance—our CNN-LSTM vs. recent FL-IDS methods.

Discussion

Mechanisms of architectural Non-IID robustness

The following mechanisms explain the observed degradation patterns. Under moderate non-IID conditions ($\alpha \geq 0.5$), degradation is minimal ($\leq 1.91\%$ from centralized). Under extreme skew ($\alpha = 0.01$), the 16.64% degradation is primarily driven by quantity imbalance, not feature heterogeneity. Four architectural mechanisms partially mitigate but do not eliminate this degradation:

- Universal Feature Learning: SHAP analysis confirms the model's reliance on universal features, including Init_Win_bytes_forward, ACK Flag Count, and Fwd Packet Length Min and TCP flags. The TCP flags used in the attack are the same across all attacks, regardless of the network, whether it is in the hospital or the factory. Port scans also utilize the same flow duration characteristics, regardless of the industry. These features are universal, so the model trained on the diverse local data distributions will tend to converge to the same relative importance order, thereby facilitating effective global aggregation.

- Hierarchical Feature Composition: Lower convolutional neural network layers recognize universal features, including TCP/IP patterns, which are invariant across client data distributions. The higher layers recognize more complex features, while the LSTM layers recognize the temporal flow characteristics. Thus, the universal features recognized by the lower layers provide the stable foundation for federated aggregation, even when the higher-layer features, which are client-specific, differ during the aggregation process.
- SMOTE Pre-conditioning: Class balancing addresses the majority class bias, which otherwise would have exacerbated the non-IID problem. Without SMOTE, minority attacks, which are present in the extreme distribution of one client, would be absent, thereby affecting global learning for these classes.
- Complementary Local Learning: Local learning on the heterogeneous data distributions results in complementary rather than conflicting learning objectives. The DDoS client learns "high-volume traffic with SYN flags → DDoS," while the Web Attack client learns "HTTP traffic with IAT patterns → Web Attack." These learning objectives are complementary, resulting in compatible gradient updates to the global model, thereby facilitating effective FedAvg aggregation without the need for proximal constraints.

The threshold of architectural robustness and the necessity of tuned aggregation

FedProx's proximal constraint is underpinned by the assumption that architectural sensitivity to client drift is caused by uncontrolled local updates, which result in incompatible model states. In contrast, the CNN-LSTM architecture used in this work has an inbuilt capability to naturally counter client drift, as updates on different clients and in different environments result in compatible states due to the use of universal feature learning. The proximal constraint effect is strongly μ-dependent: under-regularization ($\mu = 0.001$) causes catastrophic failure, while over-regularization ($\mu = 0.1$) provides the best performance under extreme skew. The corrected finding is that the architecture alone does not eliminate the need for aggregation tuning; rather, a well-tuned FedProx can complement the architecture. The implication for practitioners is that both robust architecture design AND careful aggregation hyperparameter selection are necessary for extreme non-IID scenarios.

Persistent challenges: application-layer attacks

The multi-class Precision-Recall chart analysis indicates a critical weakness in the application of flow-based intrusion detection for application-layer attacks. Although network-layer volumetric attacks show highly accurate results for Average Precision ($AP \geq 0.99$), Web Attack Brute Force and Cross-Site Scripting attacks remain challenging for the IDS model with degraded AP results of 0.81 and 0.74, respectively. Due to the fact that these attacks are carried out over legitimate HTTP or HTTPS communications, the malicious nature of these attacks is embedded entirely within the payload of the communication, such as embedded SQL commands or malicious JavaScript code. These are completely invisible to the statistical feature set extracted by CICFlowMeter and are therefore impossible to differentiate from legitimate BENIGN traffic without the application of Deep Packet Inspection techniques. In order to overcome the fundamental limitation of the application of flow-based intrusion detection models, potential avenues for future research could include: the extraction of feature sets specific to HTTP communications, such as response code ratios, request method distributions, and response header byte counts; payload-aware feature extraction techniques utilizing privacy-preserving tokenization; dynamic user profiling to identify behavioral patterns at the user session level; and the application of federated LSTM models for HTTP request sequences to identify behavioral patterns over time rather than individual feature characteristics. Furthermore, future payload-aware tokenization could be securely integrated into the FL framework utilizing privacy-preserving techniques such as federated hashing or homomorphic encryption of payload tokens allowing the network to learn critical payload semantics without exposing raw string contents to the central server.

Implications for federated learning system design

The findings of this paper have four implications for practitioners. Firstly, architectural engineering precedes algorithmic complexity, implying the importance of robust model engineering before the addition of aggregation overheads. Secondly, there should be a focus on universal features at the protocol level to support federated intrusion detection systems. Thirdly, there should be extreme heterogeneity validation (i.e., $\alpha \leq 0.01$) before the declaration of robustness under non-IID conditions. Fourthly, there should be the use of explainable artificial intelligence for generalization evaluation, where SHAP/LIME provides a framework for verifying the use of universal patterns rather than client-specific patterns. Furthermore, the lightweight nature of this architecture (~89,357 parameters) ensures minimal communication overhead per round, making it highly feasible for real-world, resource-constrained IoT deployments. Furthermore, aggregation frequency plays a critical role under extreme non-IID skew ($\alpha = 0.01$). Frequent aggregation (e.g., local epochs $E \leq 5$) bounds client drift but increases communication overhead, while infrequent aggregation ($E \geq 15$) risks irreversible weight washing. An intermediate frequency ($E = 10$) offers an optimal empirical trade-off.

Limitations and future directions

Recognized limitations: (1) Scale of federations: While this study expanded to 10-client federations, future studies will investigate massive-scale IoT networks involving hundreds of clients with various client weighting techniques. (2) The Dirichlet partitioning technique only handles skewing and does not support simultaneous multi-dimensional heterogeneity. (3) The CIC-IDS2017 dataset was obtained in a controlled environment, which might not reflect the real-world scenario. (4) Application-layer issues are architectural issues rather than federated system issues. (5) The computational efficiency of the system for resource-constrained IoT devices was not considered. Future studies will address the above five limitations. (6) Reliance on a single dataset. Simulating extreme Dirichlet skew requires massive intra-class sample depth. Cross-dataset validation across other benchmarks (e.g., TON_IoT) remains a primary trajectory for future work.

Conclusion

In this work, we perform an extensive multi-seed, multi-μ evaluation of CNN-LSTM models with embedded Explainable AI for federated IDS under extreme non-IID settings. Our centralized model reaches 96.94% accuracy (+/− 0.40%, three seeds). At extremely high heterogeneity ($\alpha = 0.01$), FedAvg provides 80.30% accuracy (+/− 0.07%) − 16.64 percentage points lower compared to our baseline, which aligns with past research on quantity-skew federated learning. In the moderately heterogeneous setting ($\alpha \geq 0.5$), nearly centralized performance is achieved with a metric of 96.22%.

Through a multi-μ sweep, we found that the FedProx model with $\mu = 0.1$ outperforms FedAvg (+4.49%) at $\alpha = 0.01$, resolving the confusion surrounding the single-μ result in our prior work. Furthermore, using multiple Explainability approaches like SHAP, LIME, t-SNE, and PCA, we find that the model uses universally discriminative features like Init_Win_bytes_forward, ACK Flag Count, Fwd Packet Length Min, and TCP Flags. Network layer attacks maintain their near-centralized F1 score (> 0.99). Application layer attacks, on the other hand, remain problematic (Web Attacks, F1 = 0.62–0.74) due to payload semantics that are beyond the scope of flows.

In summary, we demonstrate a novel set of benchmarks for federated IDS models under heterogeneous conditions. These results prove that a well-conceived federated IDS system leads to robust federated security solutions without unnecessary complexity. Our work will be fully reproducible because of providing pseudo code, mathematically defined pre-processing procedures, and XAI analyses.

Data availability

The CIC-IDS2017 dataset analyzed during the current study is publicly available from the Canadian Institute for Cybersecurity (CIC) repository at the University of New Brunswick (UNB) website: https://www.unb.ca/cic/datasets/ids-2017.html.

Received: 18 February 2026; Accepted: 8 July 2026
Published online: 15 July 2026

References

1. Gubbi, J., Buyya, R., Marusic, S. & Palaniswami, M. Internet of things (IoT): A vision, architectural elements, and future directions. *Future Gener. Comput. Syst.* **29**(7), 1645–1660. https://doi.org/10.1016/j.future.2013.01.010 (2013).
2. Statista. *Number of IoT connected devices worldwide from 2019 to 2030. Statista*. https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/ (2024).
3. Antonakakis, M., April, T., Bailey, M., Bernhard, M., Bursztein, E., Cochran, J., Durumeric, Z., Halderman, J. A., Invernizzi, L., Kallitsis, M., Kumar, D., Lever, C., Ma, Z., Mason, J., Menscher, D., Seaman, C., Sullivan, N., Thomas, K. & Zhou, Y. Understanding the Mirai botnet. In *Proceedings of the 26th USENIX Security Symposium* 1093–1110 (USENIX Association, 2017). https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis
4. Lee, R. M., Assante, M. J. & Conway, T. Analysis of the cyber attack on the Ukrainian power grid. SANS ICS. https://www.sans.org/reading-room/whitepapers/ICS/analysis-cyber-attack-ukrainian-power-grid-36183 (2016).
5. Hodo, E., Bellekens, X., Hamilton, A., Dubouilh, P.-L., Iorkyase, E., Tachtatzis, C. & Atkinson, R. Threat analysis of IoT networks using artificial neural network intrusion detection system. In *Proceedings of the 2016 International Symposium on Networks, Computers and Communications* (ISNCC) 1–6 (IEEE, 2016). https://doi.org/10.1109/ISNCC.2016.7746067
6. Voigt, P. & Von dem Bussche, A. *The EU general data protection regulation (GDPR): A practical guide* (Springer, 2017). https://doi.org/10.1007/978-3-319-57959-7.
7. Buczak, A. L. & Guven, E. A survey of data mining and machine learning methods for cyber security intrusion detection. *IEEE Commun. Surv. Tutor.* **18**(2), 1153–1176. https://doi.org/10.1109/COMST.2015.2494502 (2016).
8. Vinayakumar, R. et al. Deep learning approach for intelligent intrusion detection system. *IEEE Access* **7**, 41525–41550. https://doi.org/10.1109/ACCESS.2019.2895334 (2019).
9. McMahan, B., Moore, E., Ramage, D., Hampson, S. & y Arcas, B. A. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 54* 1273–1282 (PMLR, 2017). https://proceedings.mlr.press/v54/mcmahan17a.html
10. Yang, Q., Liu, Y., Chen, T. & Tong, Y. Federated machine learning: Concept and applications. *ACM Trans. Intell. Syst. Technol.* https://doi.org/10.1145/3298981 (2019).
11. Hard, A., Rao, K., Mathews, R., Ramaswamy, S., Beaufays, F., Augenstein, S., Eichner, H., Kiddon, C. & Ramage, D. *Federated learning for mobile keyboard prediction*. Preprint at https://arxiv.org/abs/1811.03604 (2018).
12. Doshi, R., Apthorpe, N. & Feamster, N. Machine learning DDoS detection for consumer Internet of Things devices. In *Proceedings of the 2018 IEEE Security and Privacy Workshops (SPW)* pp. 29–35. (IEEE, 2018). https://doi.org/10.1109/SPW.2018.00013
13. Ahmad, Z., Shahid Khan, A., Wai Shiang, C., Abdullah, J. & Ahmad, F. Network intrusion detection system: A systematic study of machine learning and deep learning approaches. *Trans. Emerg. Telecommun. Technol.* **32**(1), e4150. https://doi.org/10.1002/ett.4150 (2021).
14. Miorandi, D., Sicari, S., De Pellegrini, F. & Chlamtac, I. Internet of things: Vision, applications and research challenges. *Ad Hoc Netw.* **10**(7), 1497–1516. https://doi.org/10.1016/j.adhoc.2012.02.016 (2012).
15. Zhao, Y., Li, M., Lai, L., Suda, N., Civin, D. & Chandra, V. *Federated learning with non-IID data*. Preprint at https://arxiv.org/abs/1806.00582 (2018).
16. Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A. & Smith, V. Federated optimization in heterogeneous networks. In *Proceedings of Machine Learning and Systems (MLSys 2020)* **2**, pp. 429–450. https://arxiv.org/abs/1812.06127 (2020).
17. Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S. J., Stich, S. U. & Suresh, A. T. SCAFFOLD: Stochastic controlled averaging for federated learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML 2020), PMLR* **119**, pp. 5132–5143 (PMLR, 2020). https://proceedings.mlr.press/v119/karimireddy20a.html
18. Wang, J., Liu, Q., Liang, H., Joshi, G. & Poor, H. V. Tackling the objective inconsistency problem in heterogeneous federated optimization. In *Advances in Neural Information Processing Systems (NeurIPS 2020)*, **33**, pp. 7611–7623. https://proceedings.neurips.cc/paper/2020/hash/564127c03caab942e503ee6f810f54fd-Abstract.html (2020).
19. Chawla, N. V., Bowyer, K. W., Hall, L. O. & Kegelmeyer, W. P. SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **16**, 321–357. https://doi.org/10.1613/jair.953 (2002).
20. Ji, R., Kumar, N. & Padha, D. Cascading bagging and boosting ensemble methods for intrusion detection in cyber-physical systems. *Secur. Privacy* **8**, e497. https://doi.org/10.1002/spy2.497 (2025).

21. Ji, R., Padha, D., Singh, Y. & Sharma, S. Review of intrusion detection system in cyber-physical system based networks: Characteristics, industrial protocols, attacks, data sets and challenges. *Trans. Emerg. Telecommun. Technol.* **35**(9), e5029. https://doi.org/10.1002/ett.5029 (2024).
22. Ji, R., Kumar, N. & Padha, D. CNN-GWO-voting & hybrid: Ensemble learning inspired intrusion detection approaches for cyber-physical systems. *Proc. Indian Natl. Sci. Acad.* **91**, 848–862. https://doi.org/10.1007/s43538-024-00372-0 (2025).
23. Lazzarini, R., Tianfield, H. & Charissis, V. Federated learning for IoT intrusion detection. *AI* **4**(3), 509–530. https://doi.org/10.3390/ai4030028 (2023).
24. Lundberg, S. M. & Lee, S.-I. A unified approach to interpreting model predictions. In *Advances in neural information processing systems (NeurIPS 2017)*, **30**, pp. 4765–4774 https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html (2017).
25. Ribeiro, M. T., Singh, S. & Guestrin, C. "Why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2016)* 1135–1144 (ACM, 2016) https://doi.org/10.1145/2939672.2939778
26. Van der Maaten, L. & Hinton, G. Visualizing data using t-SNE. *J. Mach. Learn. Res.* **9**(86), 2579–2605 (2008).
27. Jolliffe, I. T. & Cadima, J. Principal component analysis: A review and recent developments. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* **374**(2065), Article 20150202. https://doi.org/10.1098/rsta.2015.0202 (2016).
28. Li, X., Huang, K., Yang, W., Wang, S. & Zhang, Z. On the convergence of FedAvg on non-IID data. In *Proceedings of the 8th International Conference on Learning Representations* (ICLR 2020). https://arxiv.org/abs/1907.02189 (2020).
29. Karimireddy, S. P., Jaggi, M., Kale, S., Mohri, M., Reddi, S. J., Stich, S. U. & Suresh, A. T. Breaking the centralized barrier for cross-device federated learning. In *Advances in Neural Information Processing Systems (NeurIPS 2021)*, **34**, pp. 28663–28676. https://proceedings.neurips.cc/paper/2021/hash/f0e6be4ce76ccfa73c5a540d992d0756-Abstract.html (2021).
30. Zhu, H., Xu, J., Liu, S. & Jin, Y. Federated learning on non-IID data: A survey. *Neurocomputing* **465**, 371–390. https://doi.org/10.1016/j.neucom.2021.07.098 (2021).
31. Hsieh, K., Phanishayee, A., Mutlu, O. & Gibbons, P. B. The non-IID data quagmire of decentralized machine learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML 2020), PMLR* **119**, pp. 4387–4398 (PMLR, 2020). https://proceedings.mlr.press/v119/hsieh20a.html
32. Moussaoui, J.-E., Kmiti, M., El Gholami, K. & Maleh, Y. A systematic review on hybrid AI models integrating machine learning and federated learning. *J. Cybersecur. Privacy* **5**(3), 41. https://doi.org/10.3390/jcp5030041 (2025).
33. Javaid, A., Niyaz, Q., Sun, W. & Alam, M. A deep learning approach for network intrusion detection system. In *Proceedings of the 9th EAI International Conference on Bio-inspired Information and Communications Technologies (BICT 2016)* 21–26 (ACM, 2016) https://doi.org/10.4108/eai.3-12-2015.2262516
34. Yin, C., Zhu, Y., Fei, J. & He, X. A deep learning approach for intrusion detection using recurrent neural networks. *IEEE Access* **5**, 21954–21961. https://doi.org/10.1109/ACCESS.2017.2762418 (2017).
35. Li, Z., Qin, Z., Huang, K., Yang, X. & Ye, S. Intrusion detection using convolutional neural networks for representation learning. In *Proceedings of the 24th International Conference on Neural Information Processing (ICONIP 2017), LNCS* **10638**, pp. 858–866 (Springer, 2017) https://doi.org/10.1007/978-3-319-70139-4_87
36. Nguyen, T. T. T. & Armitage, G. A survey of techniques for internet traffic classification using machine learning. *IEEE Commun. Surv. Tutor.* **10**(4), 56–76. https://doi.org/10.1109/SURV.2008.080406 (2008).
37. Zhang, C., Costa-Perez, X. & Patras, P. Enabling graph neural network-based intrusion detection in IoT networks. *IEEE Internet Things J.* **10**(11), 9321–9335. https://doi.org/10.1109/JIOT.2023.3244098 (2023).
38. Wang, M., Zheng, K., Yang, Y. & Wang, X. An explainable attention network for intrusion detection. *IEEE Trans. Dependable Secure Comput.* **19**(6), 3847–3861. https://doi.org/10.1109/TDSC.2021.3126359 (2022).
39. Lin, X., Xiong, G., Gou, G., Li, Z., Shi, J. & Yu, J. ET-BERT: A contextualized datagram representation with pre-training transformers for encrypted traffic classification. In *Proceedings of the ACM Web Conference 2022 (WWW '22)* 633–642 (ACM, 2022) https://doi.org/10.1145/3485447.3512217
40. Nguyen, T. D., Marchal, S., Miettinen, M., Fereidooni, H., Asokan, N. & Sadeghi, A.-R. DIoT: A federated self-learning anomaly detection system for IoT. In *Proceedings of the 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS)* 756–767 (IEEE, 2019) https://doi.org/10.1109/ICDCS.2019.00080
41. Zhao, R., Yin, Y., Shi, Y. & Xue, Z. Intelligent intrusion detection based on federated learning aided long short-term memory. *Phys. Commun.* **42**, 101157. https://doi.org/10.1016/j.phycom.2020.101157 (2020).
42. Liu, Y., Ma, Z., Liu, X., Ma, S. & Ren, K. Boosting privately: Privacy-preserving federated extreme boosting for mobile crowdsensing. In *Proceedings of the 41st IEEE International Conference on Distributed Computing Systems (ICDCS 2021)* 1–11 (IEEE, 2021) https://doi.org/10.1109/ICDCS51616.2021.00059
43. Preuveneers, D. et al. Chained anomaly detection models for federated learning: An intrusion detection case study. *Appl. Sci.* **8**(12), 2663. https://doi.org/10.3390/app8122663 (2018).
44. Ahmed, U., Jiangbin, Z., Khan, S. & Sadiq, M. T. Consensus hybrid ensemble machine learning for intrusion detection with explainable AI. *J. Netw. Comput. Appl.* **235**, 104091. https://doi.org/10.1016/j.jnca.2024.104091 (2025).
45. Ahmed, U. et al. Hybrid bagging and boosting with SHAP based feature selection for enhanced predictive modeling in intrusion detection systems. *Sci. Rep.* **14**, 30532. https://doi.org/10.1038/s41598-024-81151-1 (2024).
46. Tavallaee, M., Bagheri, E., Lu, W. & Ghorbani, A. A. A detailed analysis of the KDD CUP 99 data set. In *Proceedings of the 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)* 1–6 (IEEE, 2009) https://doi.org/10.1109/CISDA.2009.5356528
47. Lashkari, A. H., Draper Gil, G., Mamun, M. S. I. & Ghorbani, A. A. Characterization of Tor traffic using time based features. In *Proceedings of the 3rd International Conference on Information Systems Security and Privacy (ICISSP 2017)* 253–262 (SciTePress, 2017). https://doi.org/10.5220/0006105602530262
48. Khudhair, R. T., Hock, G. C. & Abu-Bakar, A. B. A comparative analysis of federated learning on non-IID data for an intrusion detection system. *Emerg. Trends Eng. Sustain.* **1**(1), 37–44. https://doi.org/10.65505/ETES-25-0011 (2025).
49. Khishe, M. & Mosavi, M. R. Chimp optimization algorithm. *Expert Syst. Appl.* **149**, 113338. https://doi.org/10.1016/j.eswa.2020.113338 (2020).
50. Wang, W. & Liu, Z. Design and implementation of network intrusion detection system (NIDS) based on deep learning. *Secur. Privacy* https://doi.org/10.1002/spy2.70133 (2025).
51. Fatema, K. et al. Federated XAI IDS: An explainable and safeguarding privacy approach to detect intrusion combining federated learning and SHAP. *Future Internet* **17**(6), 234. https://doi.org/10.3390/fi17060234 (2025).
52. Rajagopalan, N. Federated learning and explainable AI-driven intrusion detection with hyperband optimization. *J. Comput. Virol. Hack. Tech.* **21**, 28. https://doi.org/10.1007/s11416-025-00571-3 (2025).
53. AlHayan, A. & Al-Muhtadi, J. Federated learning-powered real-time behavioral intrusion detection leveraging LSTM, attention, GANs, and large language models. *Sci. Rep.* **16**(1), 10172 (2026).
54. Beutel, D. J., Topal, T., Mathur, A., Qiu, X., Parcollet, T. & Lane, N. D. Flower: A friendly federated learning research framework. Preprint at https://doi.org/10.48550/arXiv.2007.14390 (2020).
55. Bahdanau, D., Cho, K. & Bengio, Y. *Neural machine translation by jointly learning to align and translate*. Preprint at arXiv:1409.0473 (2014).

56. Halbouni, A. et al. CNN-LSTM: Hybrid deep neural network for network intrusion detection system. *IEEE Access* **10**, 99837–99849. https://doi.org/10.1109/ACCESS.2022.3206425 (2022).
57. Sumathi, S., Rajesh, R. & Lim, S. Intelligent intrusion detection system for IoT networks: A feature enhanced deep belief network approach. *IEEE Internet Things J.* **13**(3), 4477–4491. https://doi.org/10.1109/JIOT.2025.3633297 (2026).
58. Sumathi, S. & Rajesh, R. A dynamic BPN-MLP neural network DDoS detection model using hybrid swarm intelligent framework. *Indian J. Sci. Technol.* **16**(43), 3890–3904 (2023).
59. Sharafaldin, I., Habibi Lashkari, A. & Ghorbani, A. A. Toward generating a new intrusion detection dataset and intrusion traffic characterization. *Int. Conf. Inf. Syst. Secur. Privacy* https://doi.org/10.5220/0006639801080116 (2018).

Acknowledgements
The authors would like to acknowledge the support of the LaSTI Laboratory at ENSA Khouribga, Sultan Moulay Slimane University, for providing access to academic resources and guidance throughout the research process. We also thank our colleagues for their constructive feedback during the manuscript revision phase.

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Conceptualization, J.-E.M., M.K. and Y.M.; methodology, J.-E.M., M.K. and Y.M.; software, J.-E.M.; validation, Y.M. and K.E.G.; formal analysis, J.-E.M.; investigation, J.-E.M.; resources, J.-E.M.; data curation, J.-E.M. and M.K.; writing—original draft preparation, J.-E.M. and Y.M.; writing—review and editing, J.-E.M., M.K., Y.M. and K.E.G.; visualization, J.-E.M.; supervision, Y.M.; project administration, Y.M.; funding acquisition, Y.M. All authors have read and agreed to the published version of the manuscript.

Funding
Not applicable.

Declarations

Competing interests
The authors declare that they have no competing interests.

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